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Gabelberger et al.

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(54) **BONE PLATE, BONE PLATE SYSTEM, AND METHOD OF USING THE SAME**

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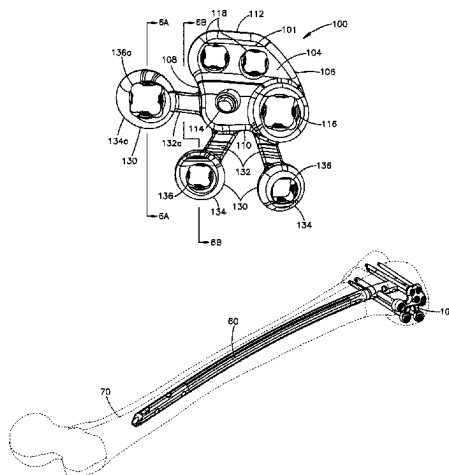
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(57) **ABSTRACT**
A bone plate comprises a plate body and at least one tab. The plate body defines an inner body surface configured to face an underlying bone, an outer body surface opposite the inner body surface, and an outer side surface that extends between the inner body surface and the outer body surface. The outer side surface defines an outer perimeter of the plate body. The at least one tab includes a head and an arm that extends from the plate body to the head. The tab defines a tab aperture that extends through the head and is configured to receive a bone anchor. The bone anchor can be coupled with a screw hole in the nail. The arm is configured to deflect with respect to
(Continued)



the plate body so as to move the head between a pre-fixation position and a fixation position spaced from the pre-fixation position.

20 Claims, 18 Drawing Sheets

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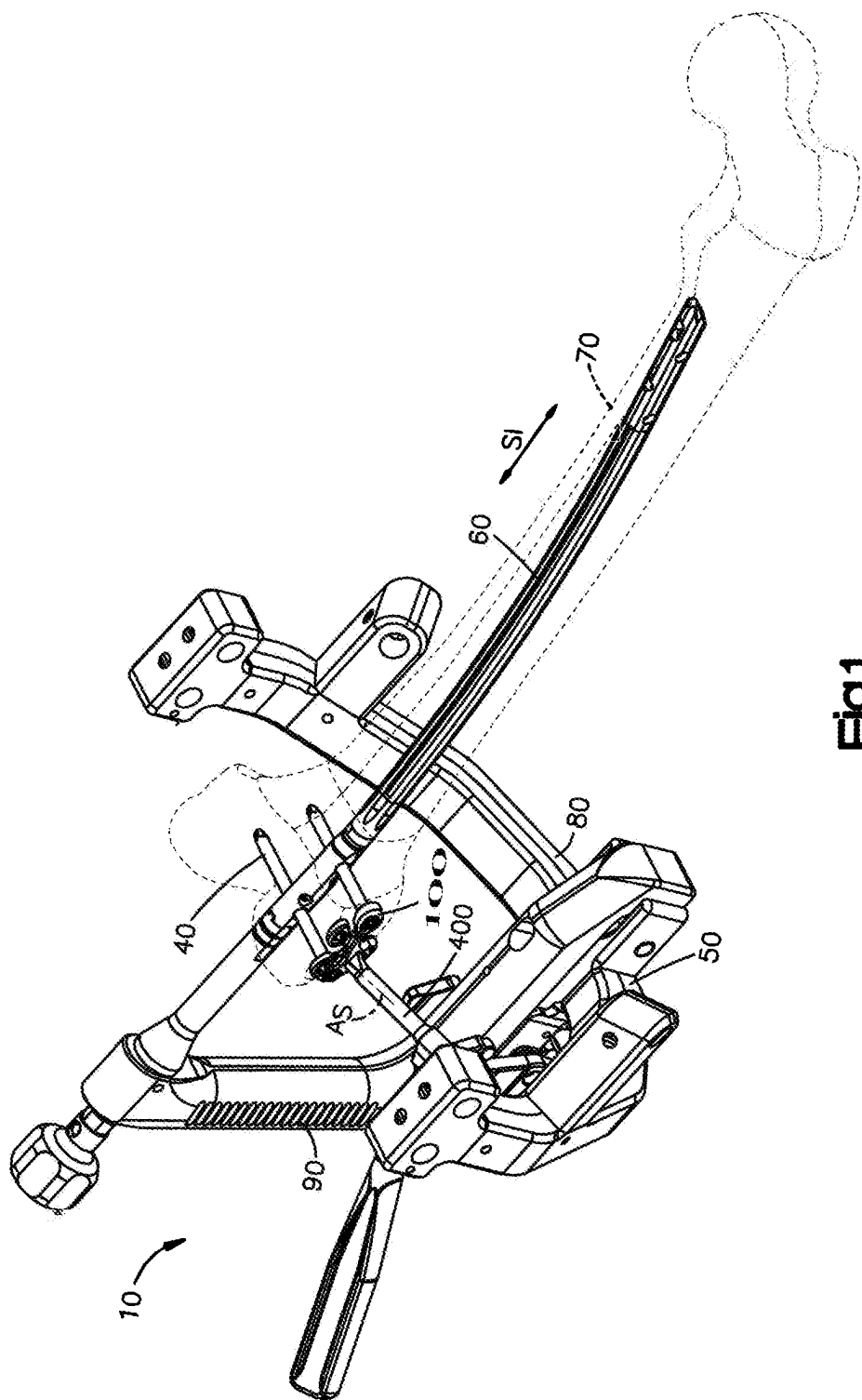
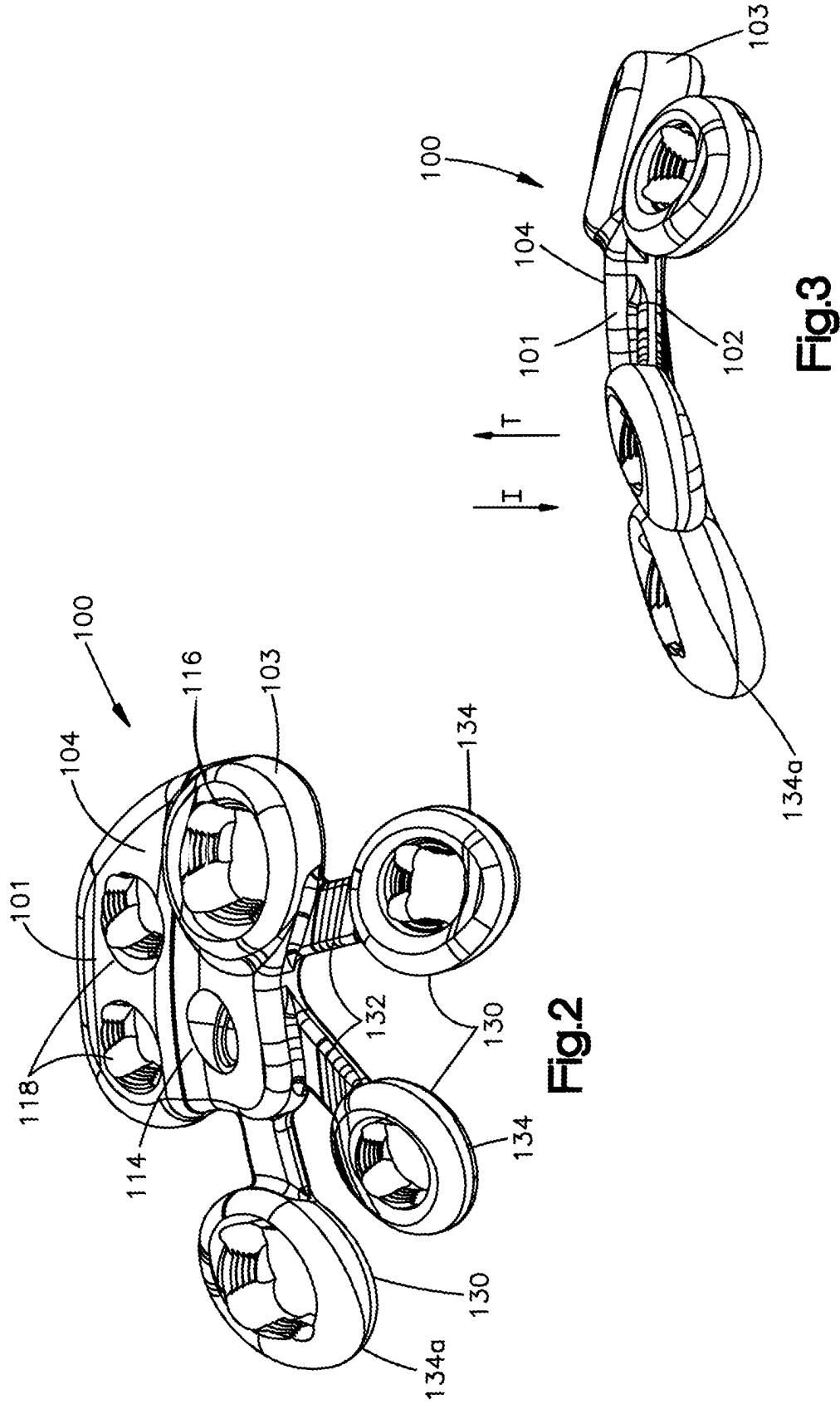
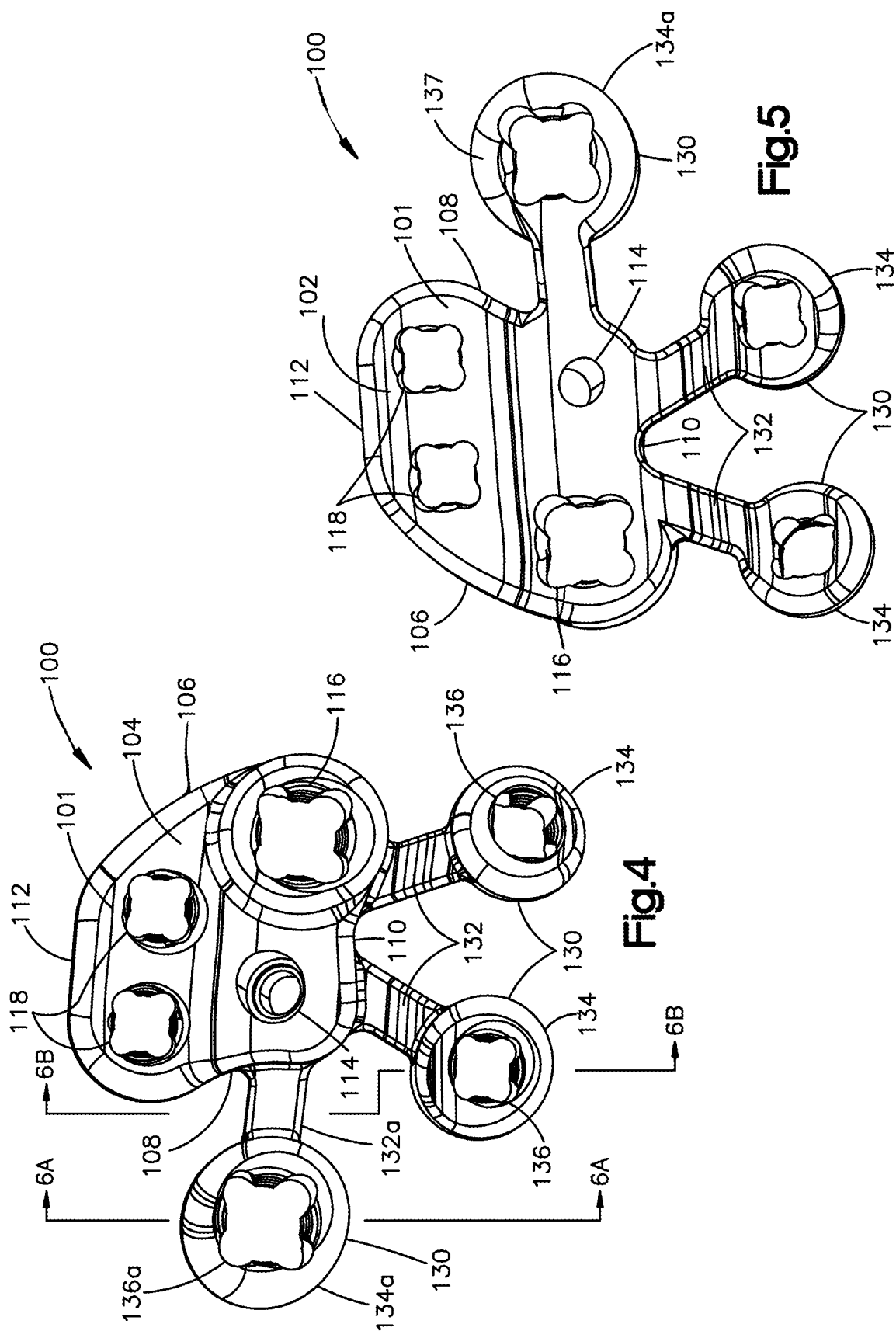
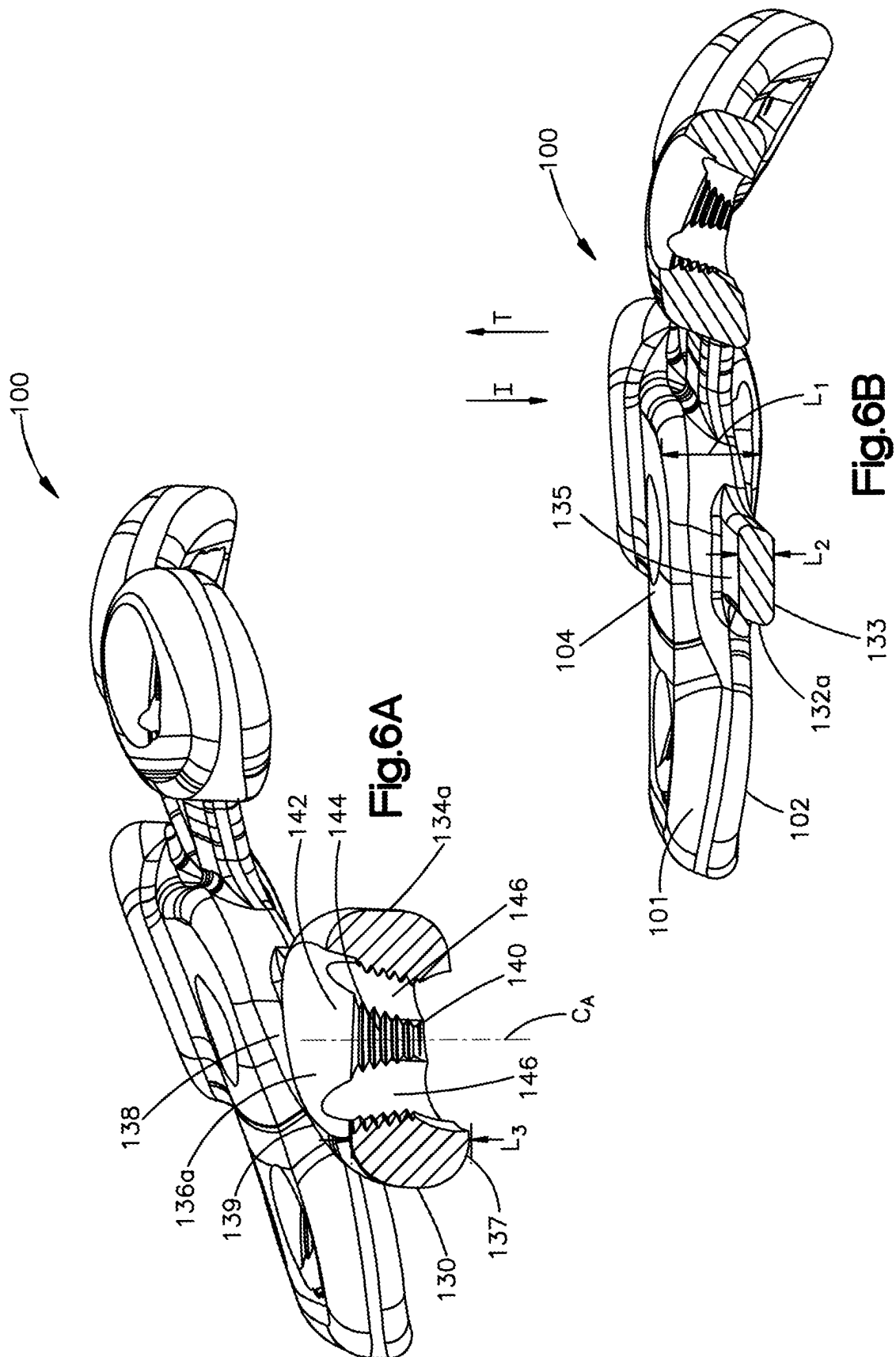
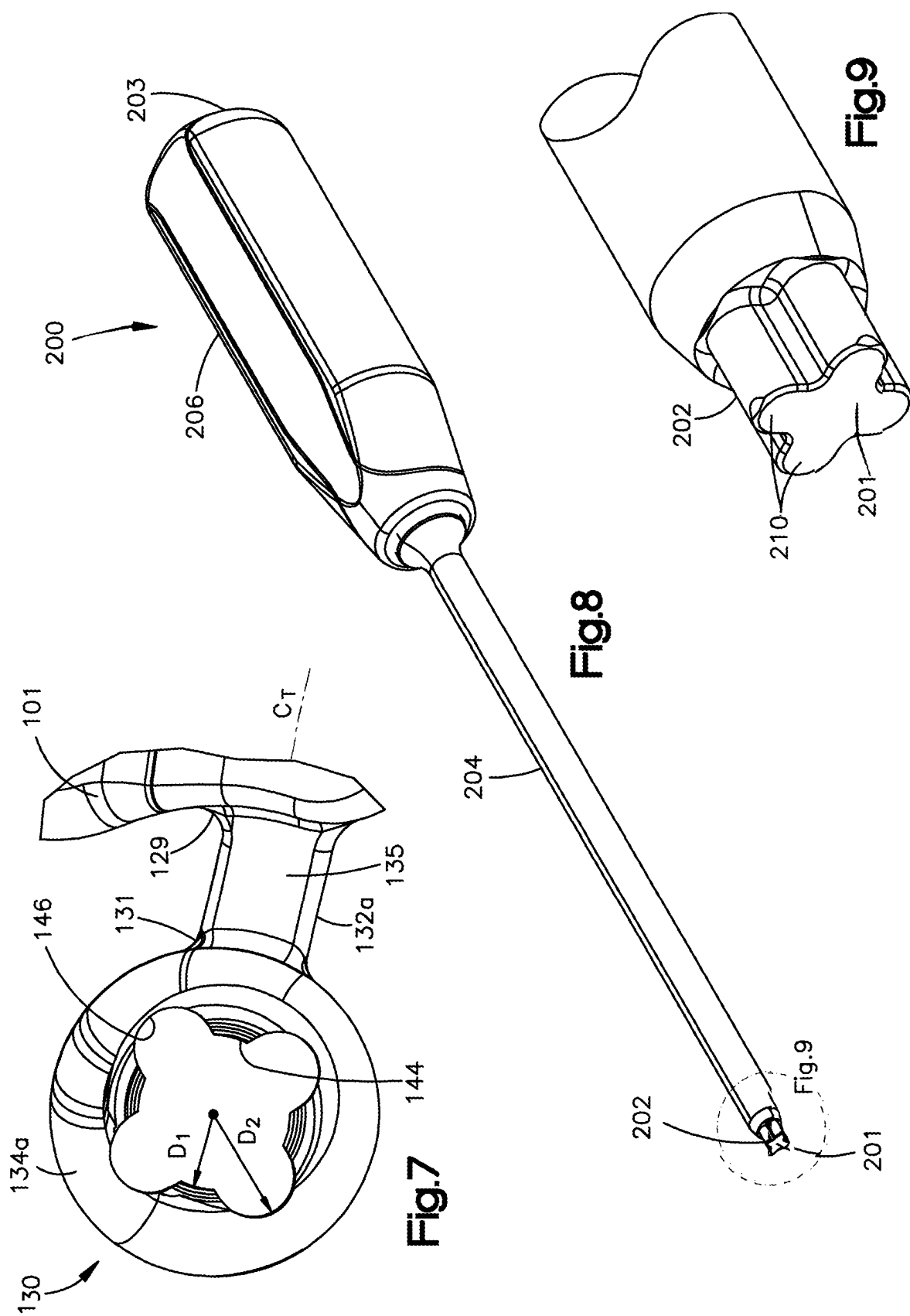


Fig.1









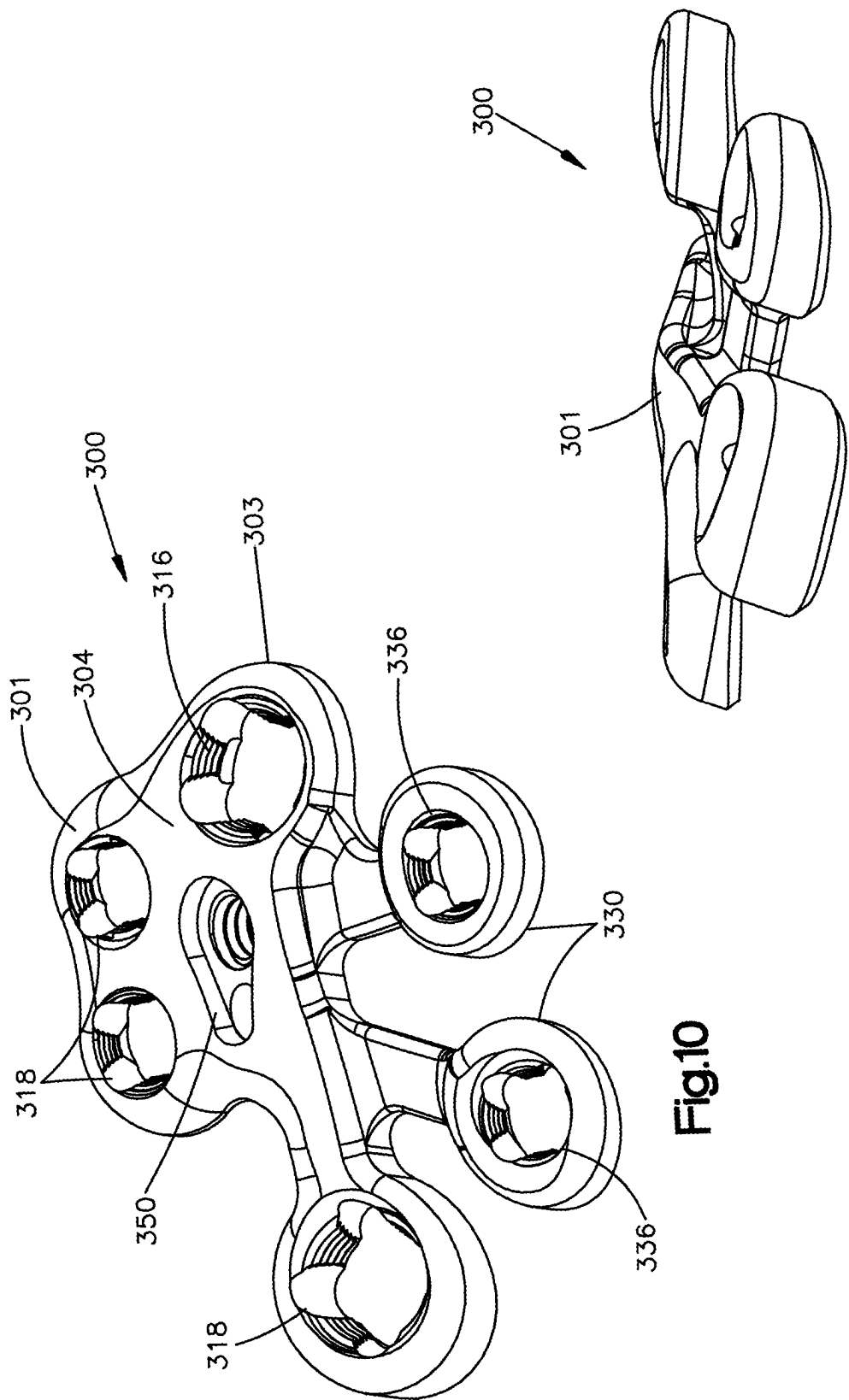
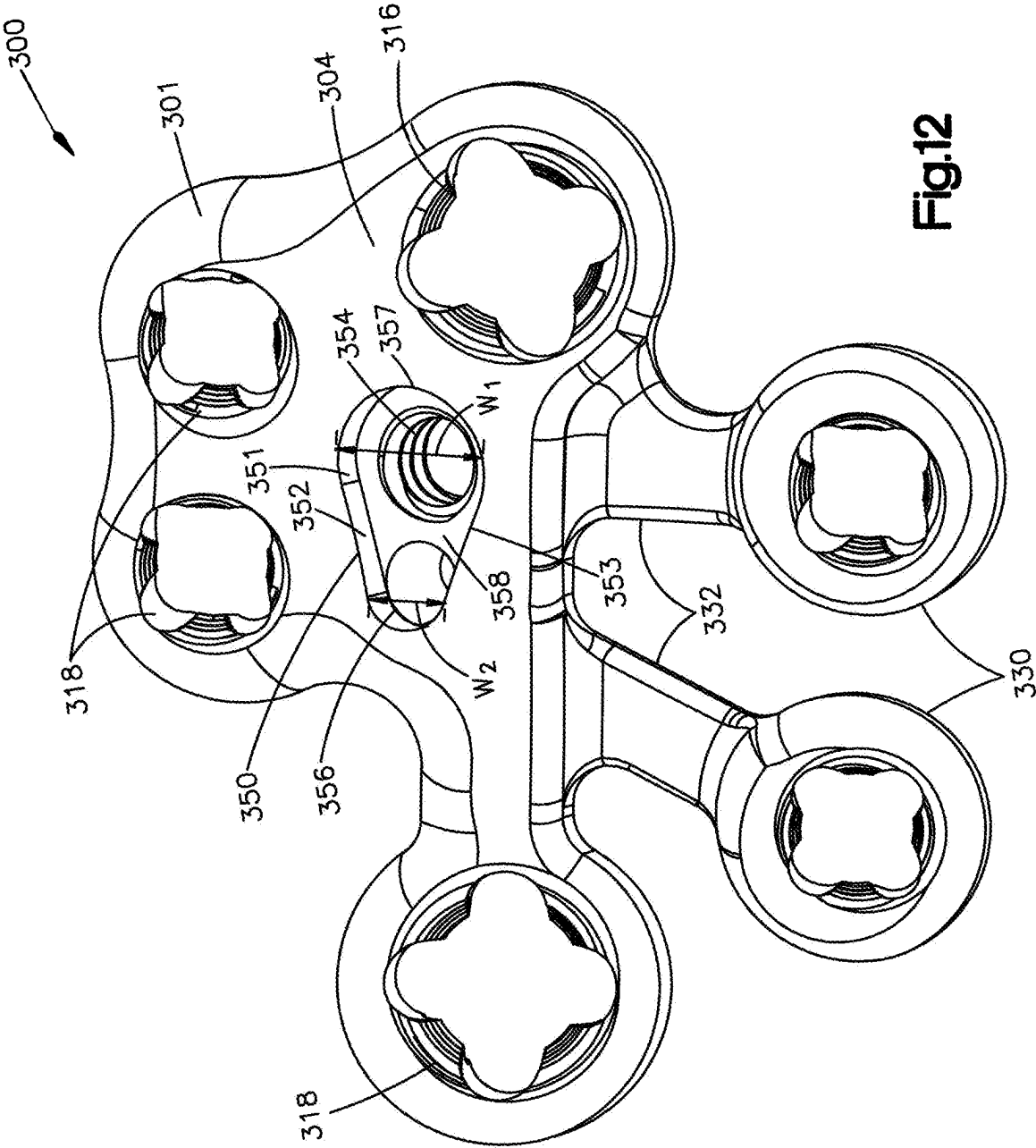
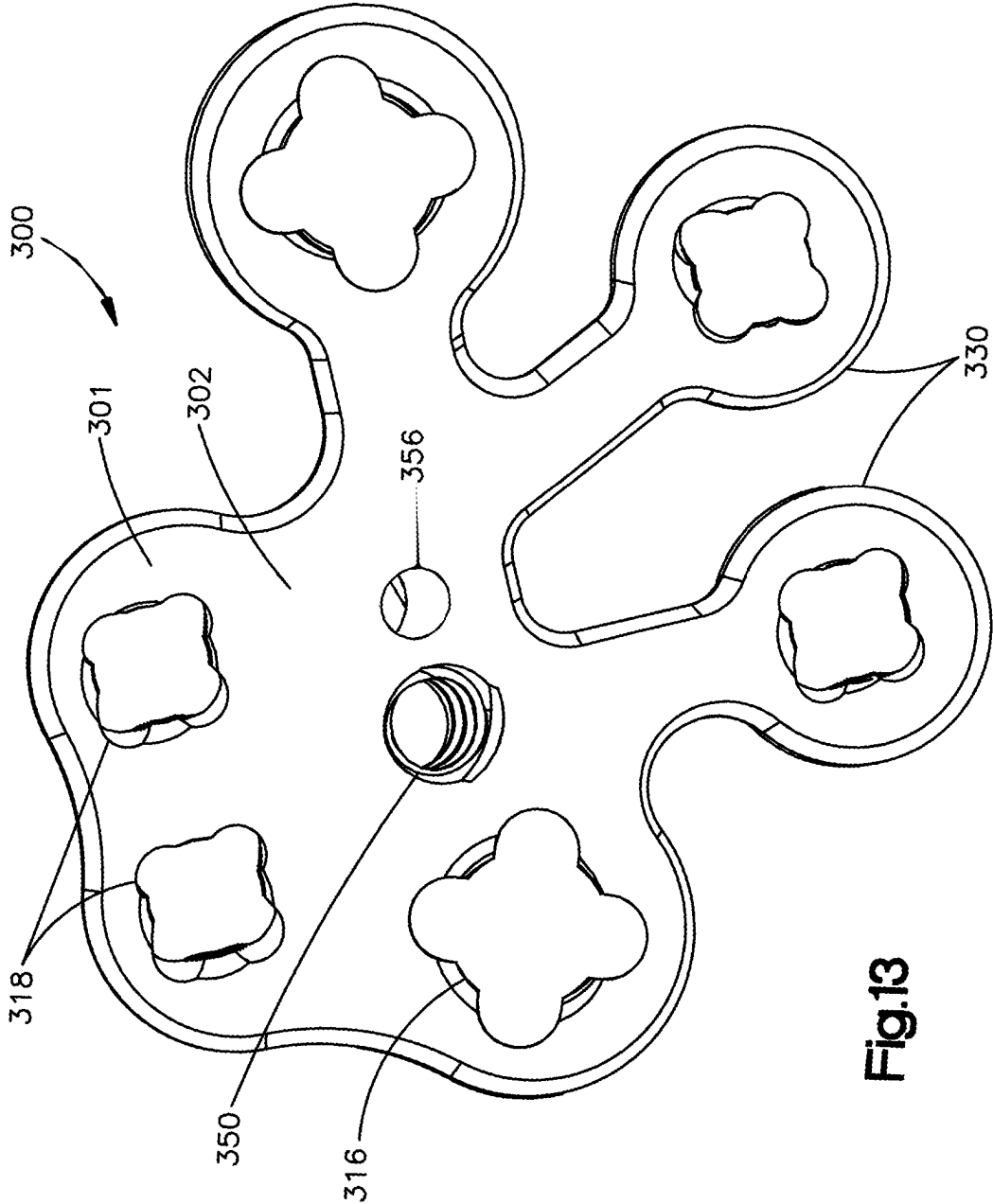
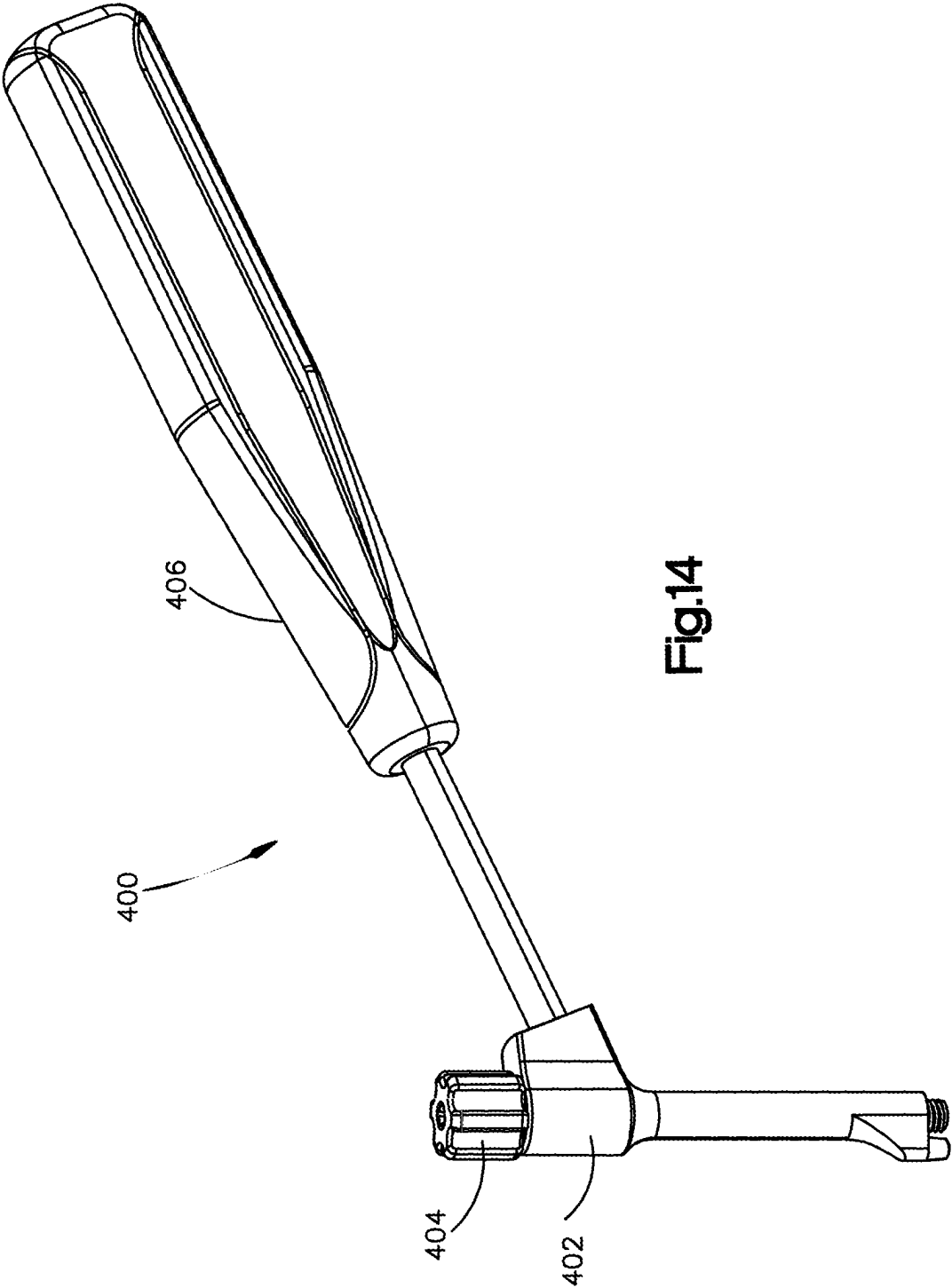


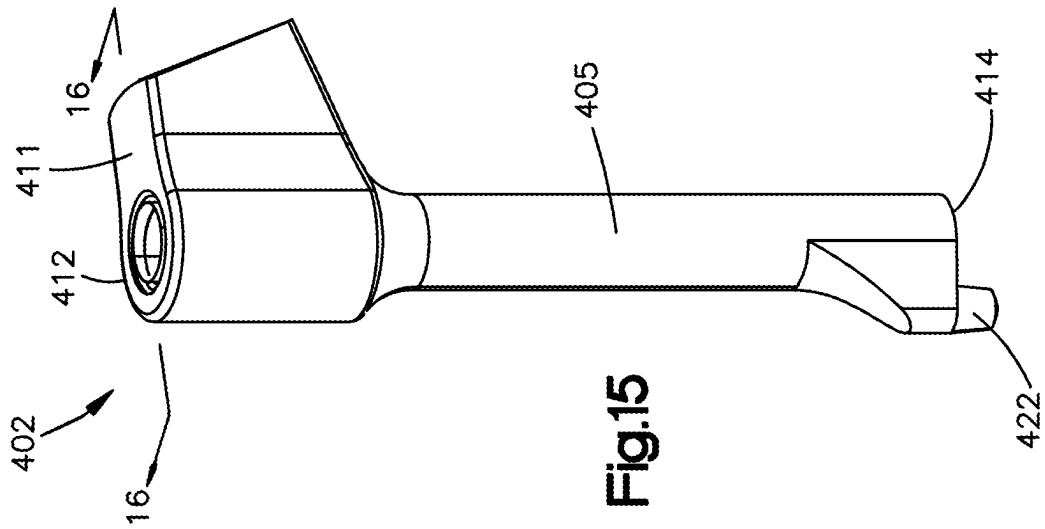
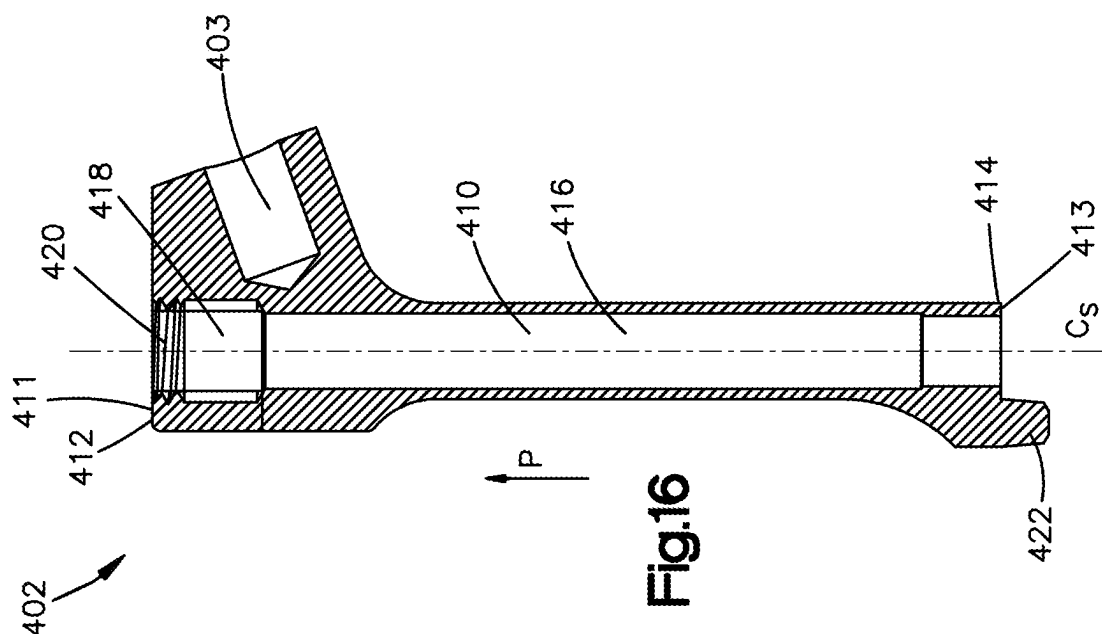
Fig.10

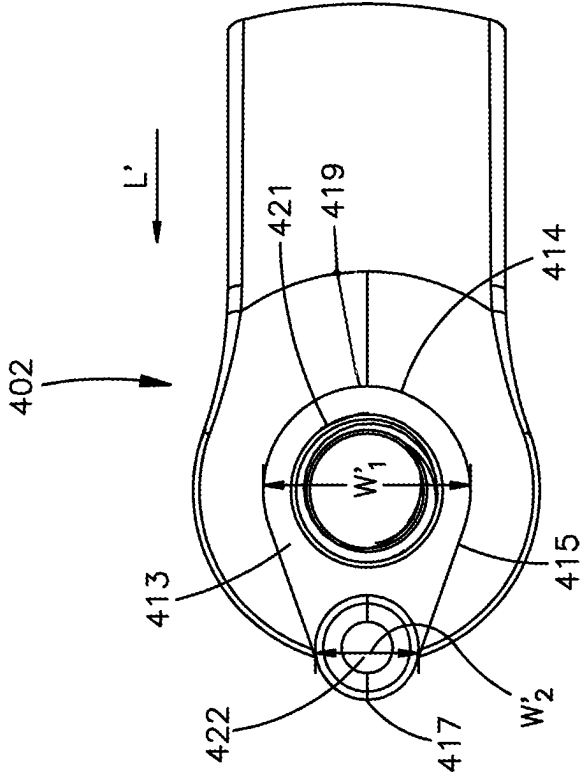
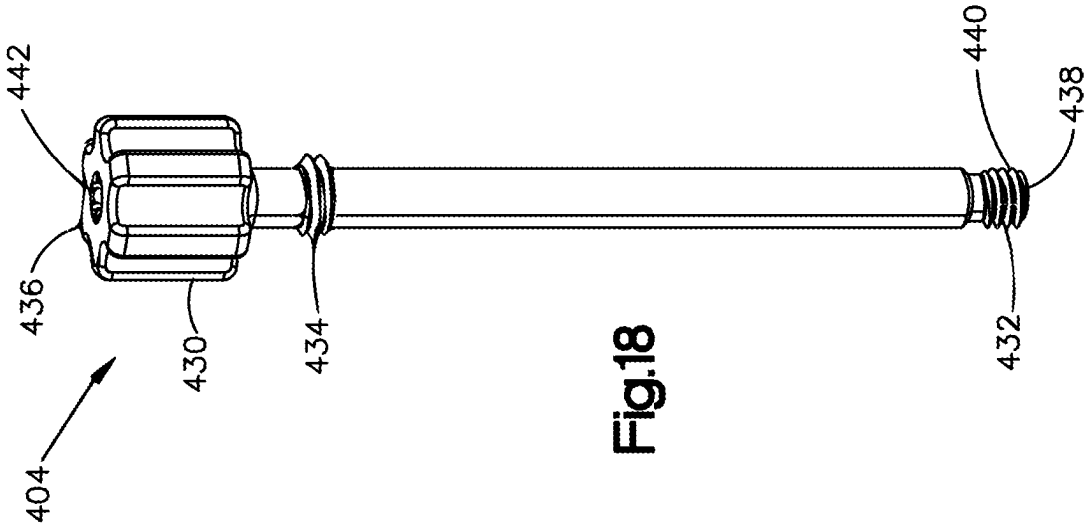
Fig.11

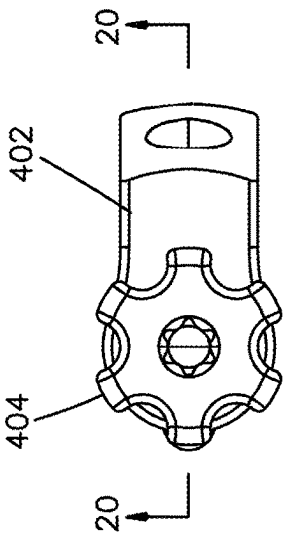
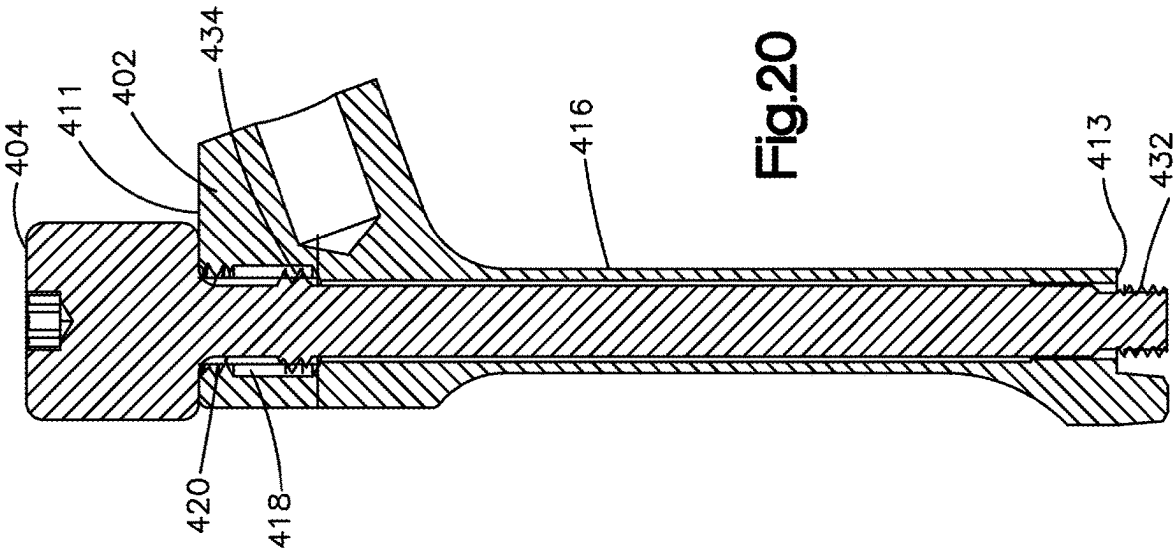


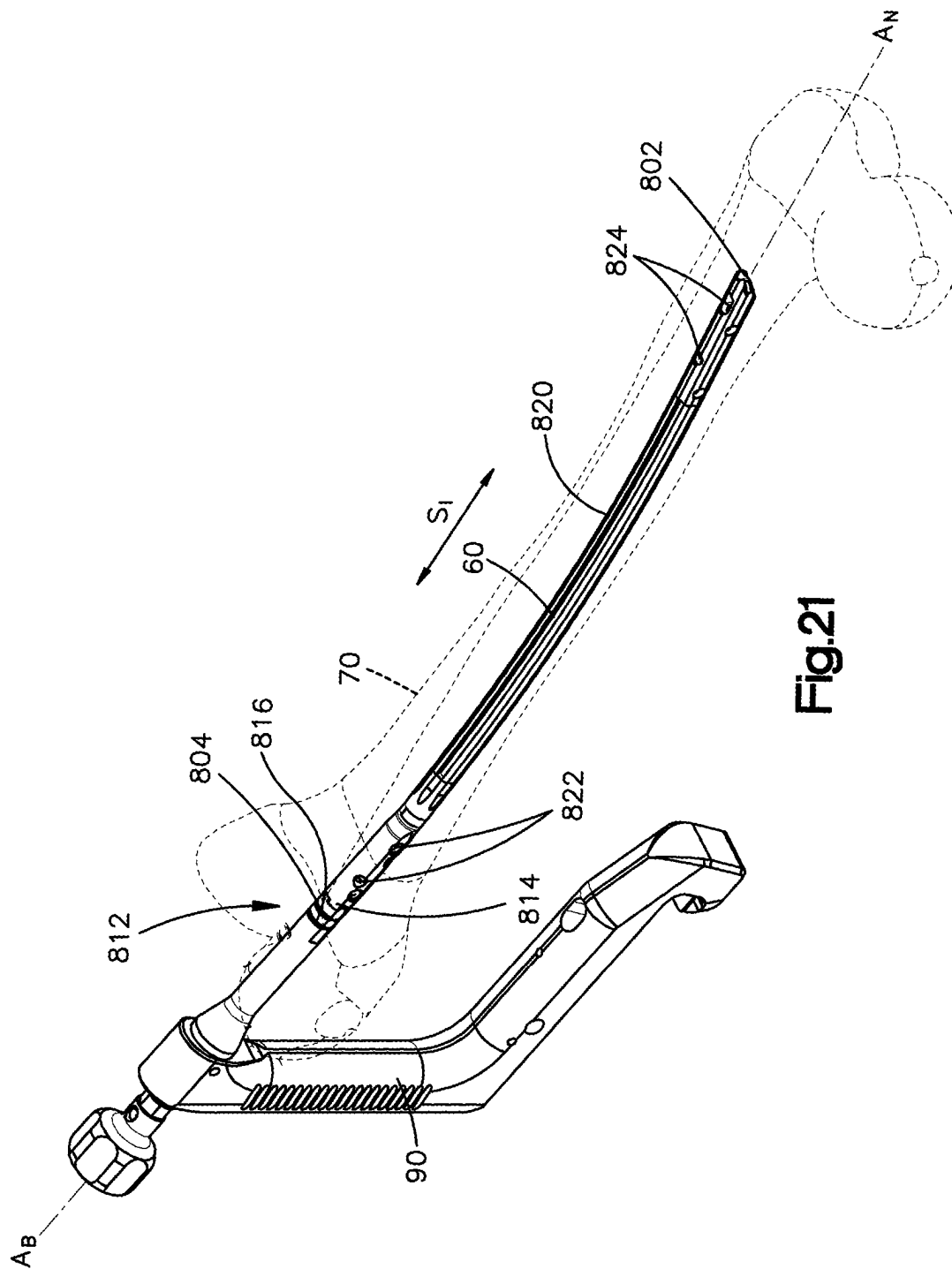


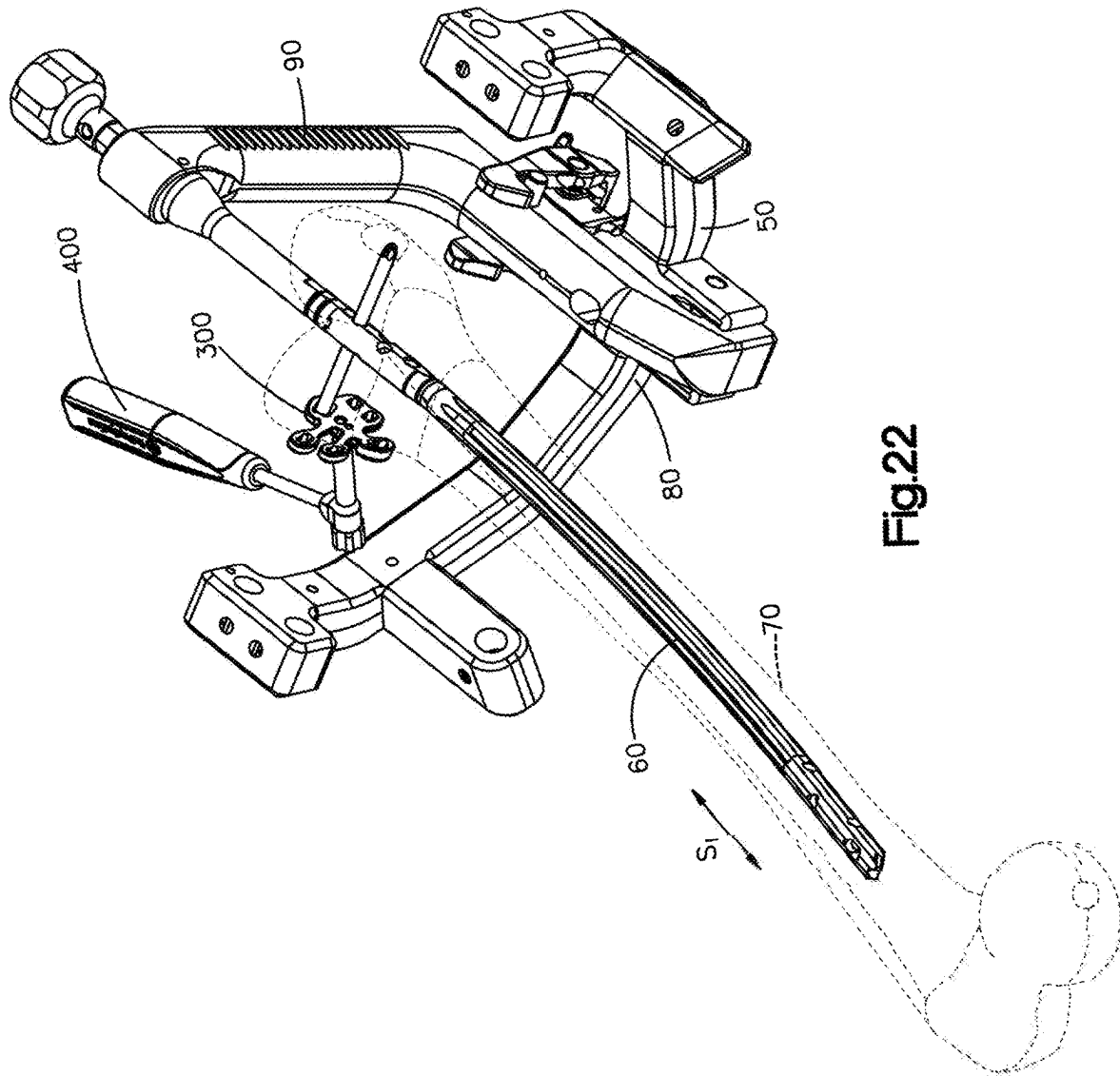












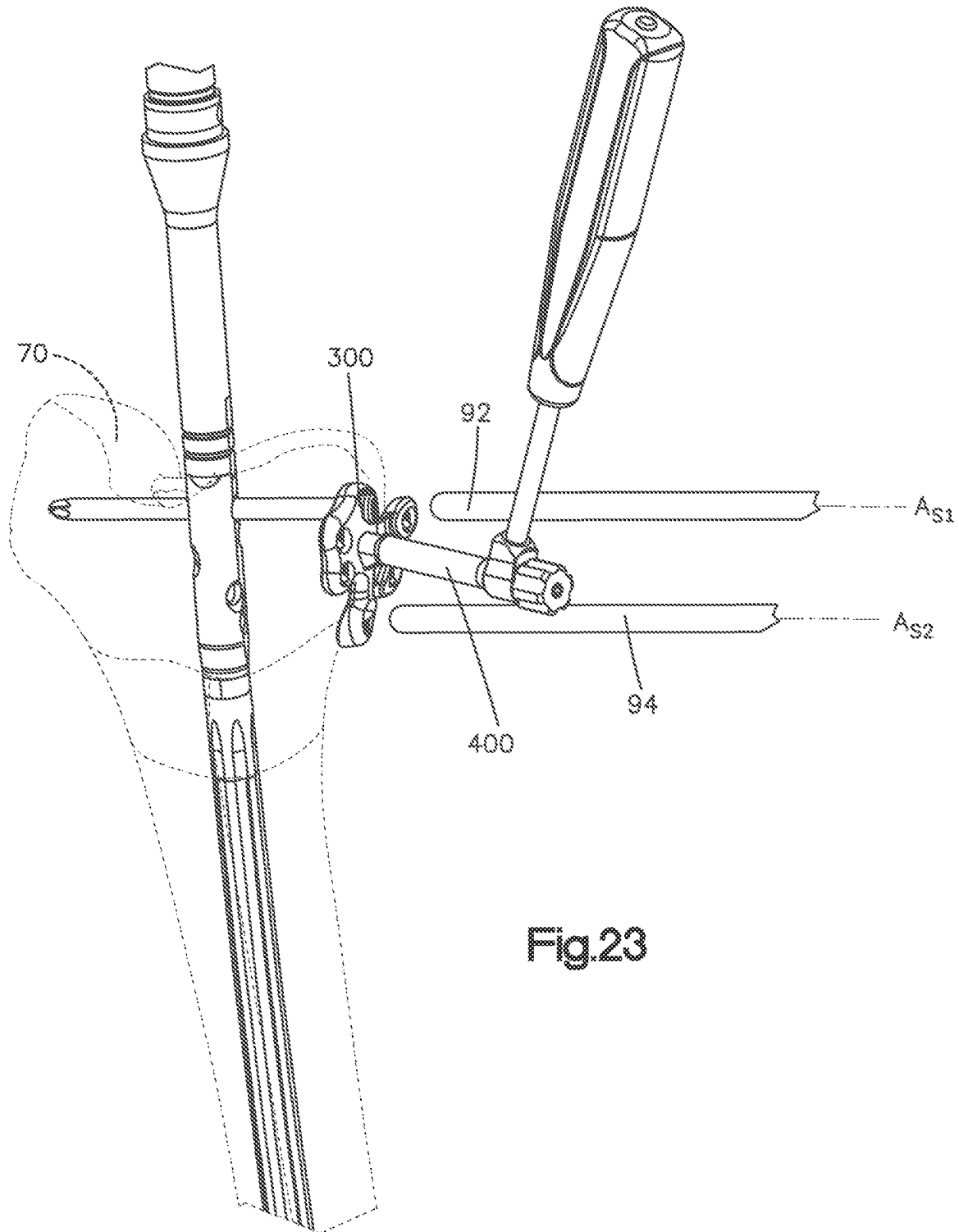


Fig.23

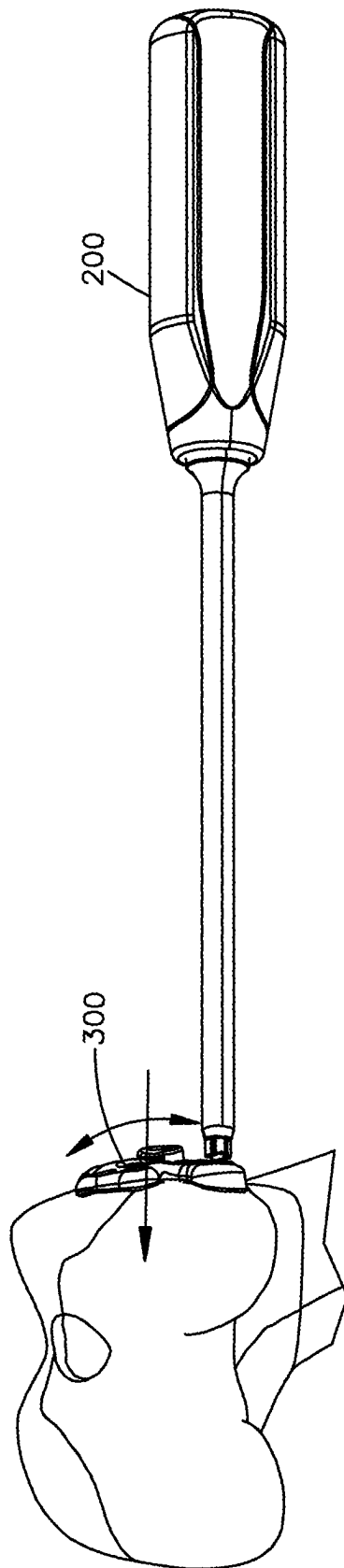
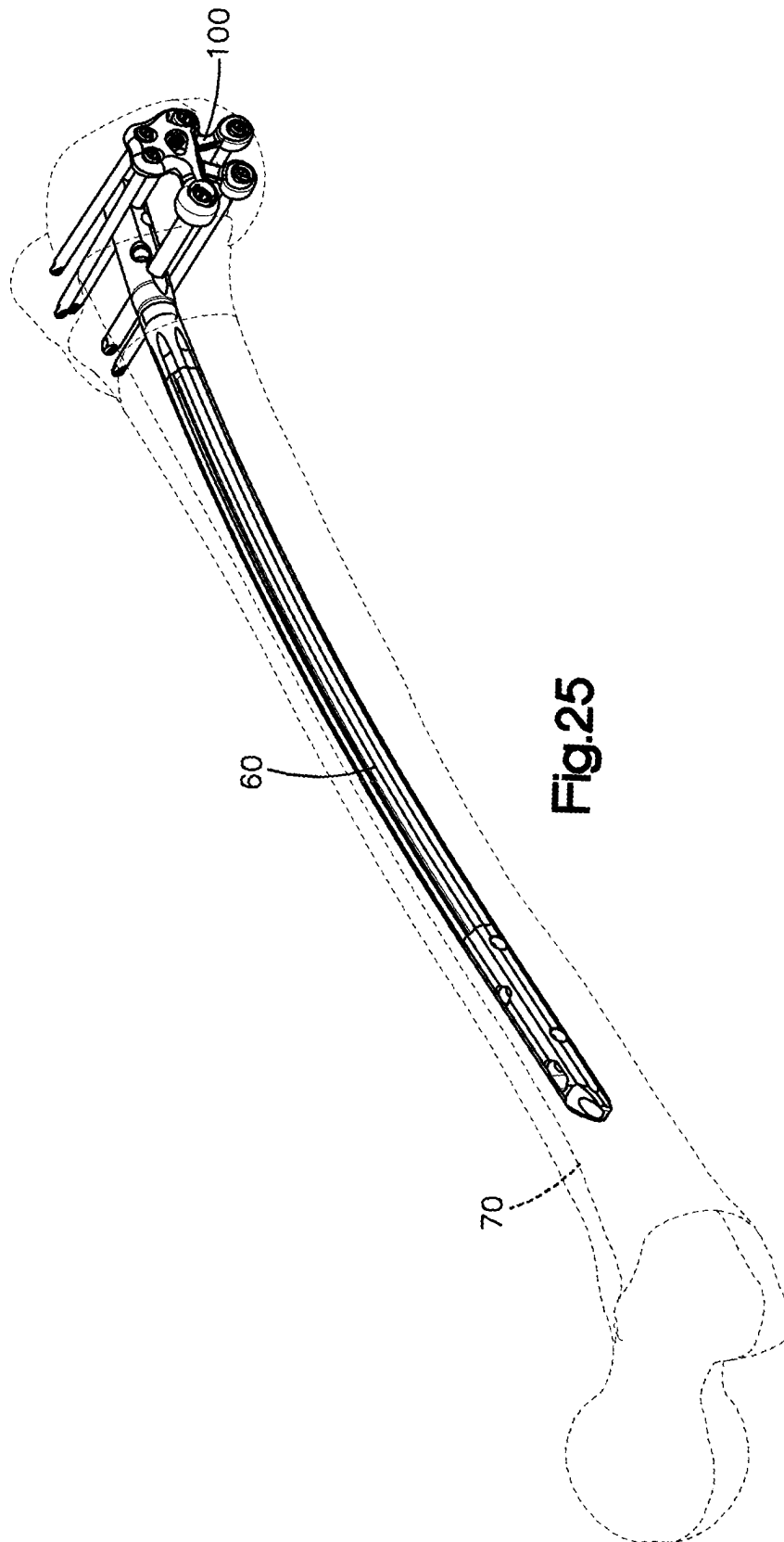
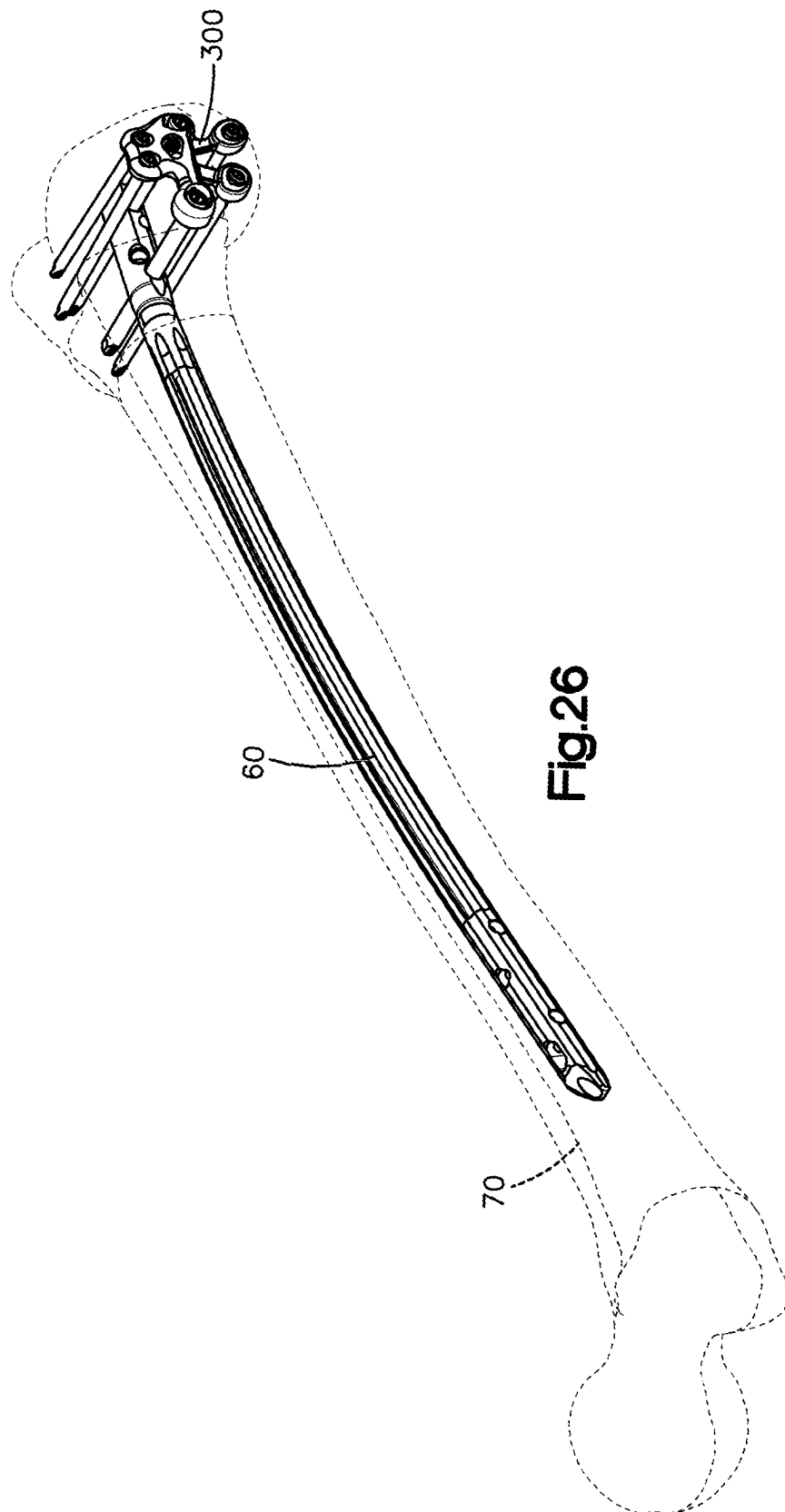


Fig.24





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**BONE PLATE, BONE PLATE SYSTEM, AND
METHOD OF USING THE SAME****CROSS-REFERENCE TO RELATED
APPLICATIONS**

This is a continuation of U.S. patent application Ser. No. 18/060,169 filed Nov. 30, 2022, which is a continuation of U.S. patent application Ser. No. 17/071,669 filed Oct. 15, 2020, the disclosure of each of which is hereby incorporated by reference as if set forth in its entirety herein.

TECHNICAL FIELD

The present disclosure relates to systems, kits, assemblies, and methods for the placement and fixation of a bone plate against a bone for attachment to an intramedullary nail in a medullary canal of the bone.

BACKGROUND

Intramedullary nails have long been used to treat fractures in long bones of the body such as fractures in femurs, tibias, and humeri. To treat such fractures, the intramedullary nail is inserted into a medullary canal of the long bone such that the nail extends across one or more fractures in the long bone to segments of the long bone that are separated by the one or more fractures. Bone anchors are then inserted through the bone and into the intramedullary nail on opposing sides of the fracture, thereby fixing the intramedullary nail to the bone. The intramedullary nail can remain in the medullary canal at least until the fracture is fused.

The foregoing background discussion is intended solely to aid the reader. It is not intended to limit the innovations described herein. Thus, the foregoing discussion should not be taken to indicate that any particular element of a prior system is unsuitable for use with the innovations described herein, nor is it intended to indicate that any element is essential in implementing the innovations described herein.

SUMMARY

The foregoing needs are met, to a great extent, by the system and method disclosed in the present application.

Retrograde femoral nailing is often used for fixation of far distal fractures of the femur. When this occurs, often in combination with poor bone quality and/or in periprosthetic settings, additional bone fixation may be required. The proposed system links a small plate or washer to the distal end of the nail by means of a plurality of bone screws. In addition, there will be a plurality of peripheral bone screws that engage more bone than the nail's bone screws alone would, for improved fixation. Proper placement of the washer may present difficulty without the aid of an appropriate holding instrument. Attachment of a holding instrument may present difficulty to operating personnel, and/or the connection may not be rigid enough.

The disclosure relates to a bone plate comprising a plate body and at least one tab. The plate body defines an inner body surface configured to face an underlying bone, an outer body surface opposite the inner body surface, and an outer side surface that extends between the inner body surface and the outer body surface. The outer side surface defines an outer perimeter of the plate body. The at least one tab includes a head and an arm that extends from the plate body to the head. The tab defines a tab aperture that extends through the head and is configured to receive a bone anchor

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body. The arm is configured to deflect with respect to the plate body so as to move the head between a pre-fixation position and a fixation position spaced from the pre-fixation position.

According to another aspect of the present disclosure, a surgical kit for securing the bone plate to a bone is provided. The surgical kit comprises the bone plate and a bone plate adjustment tool. The bone plate adjustment tool is configured to be received within the tab aperture of the at least one tab member to transition the at least one tab member from the pre-fixation position to the fixation position.

According to another aspect of the present disclosure, a bone plate holder for holding a bone plate during a surgical procedure is disclosed. The bone plate holder comprises a body and a shaft. The body defines a channel that extends through the body from a proximal surface of the body to a distal surface of the body. The shaft has a distal end that includes a coupling element configured to couple the shaft to the bone plate, and the shaft further having a proximal end opposite the distal end, wherein the proximal end includes a control element configured to control the coupling element to couple to the bone plate. The shaft is configured to extend through the channel, such that the coupling element extends out from the distal surface of the body.

This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description section. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not constrained to limitations that solve any or all disadvantages noted in any part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing summary, as well as the following detailed description of illustrative embodiments of the implant of the present application, will be better understood when read in conjunction with the appended drawings. For the purposes of illustrating the implants of the present application, there is shown in the drawings illustrative embodiments. It should be understood, however, that the application is not limited to the precise arrangements and instrumentalities shown. In the drawings:

FIG. 1 illustrates a perspective view of a system according to one aspect having a bone plate holder supported and an aiming assembly that is attached to an intramedullary nail received in a medullary canal of a bone;

FIG. 2 illustrates a top perspective view of a bone plate, according to an aspect of this disclosure;

FIG. 3 illustrates a view of a side of the bone plate shown in FIG. 2;

FIG. 4 illustrates a top view of the bone plate shown in FIG. 2;

FIG. 5 illustrates a bottom view of the bone plate shown in FIG. 2;

FIG. 6A illustrates a side cross-section view of the bone plate taken along line 6A-6A in FIG. 4;

FIG. 6B illustrates a side cross-section view of the bone plate taken along line 6B-6B in FIG. 4;

FIG. 7 illustrates a close-up top view of at least one tab member of the bone plate shown in FIG. 2;

FIG. 8 illustrates a perspective view of a bone plate adjustment tool, according to an aspect of this disclosure;

FIG. 9 illustrates a close-up view of a distal end of the bone plate adjustment tool shown in FIG. 8;

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FIG. 10 illustrates a top perspective view of another aspect of a bone plate, according to an aspect of this disclosure;

FIG. 11 illustrates a view of a side of the bone plate shown in FIG. 10;

FIG. 12 illustrates a top view of the bone plate shown in FIG. 10;

FIG. 13 illustrates a bottom view of the bone plate shown in FIG. 10;

FIG. 14 illustrates a perspective view of a bone plate holder, according to an aspect of this disclosure;

FIG. 15 illustrates a perspective view of a body of the bone plate holder shown in FIG. 14;

FIG. 16 illustrates a cross-sectional view of the bone plate holder shown in FIG. 15 taken along line 16-16;

FIG. 17 illustrates a bottom view of the bone plate holder shown in FIG. 14;

FIG. 18 illustrates a perspective view of a shaft of the bone plate holder shown in FIG. 14;

FIG. 19 illustrates a top view of the shaft within the body of the bone plate holder shown in FIG. 14;

FIG. 20 illustrates a cross-sectional view of the shaft within the body shown in FIG. 19 taken along line 20-20;

FIG. 21 illustrates a perspective view of an intramedullary nail that is received in medullary canal of a bone, the intramedullary nail coupled to a handle shown in FIG. 1;

FIG. 22 illustrates a perspective view of a system comprising the intramedullary nail and handle as shown in FIG. 21 with an aiming guide and the bone plate holder shown in FIG. 14 fastened to the bone plate shown in FIG. 10;

FIG. 23 illustrates a side view of the system of FIG. 22 with the bone plate holder supporting the bone plate and a pair of guide sleeves;

FIG. 24 illustrates a side view of the bone plate shown in FIG. 10 on a bone with the bone plate adjustment tool shown in FIG. 8 inserted into one of the tabs of the bone plate;

FIG. 25 illustrates a perspective view of the bone plate shown in FIG. 2 fastened to the intramedullary nail; and

FIG. 26 illustrates a perspective view of the bone plate shown in FIG. 10 fastened to the intramedullary nail.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

The present disclosure can be understood more readily by reference to the following detailed description taken in connection with the accompanying figures and examples, which form a part of this disclosure. It is to be understood that this disclosure is not limited to the specific devices, methods, applications, conditions or parameters described and/or shown herein, and that the terminology used herein is for the purpose of describing particular embodiments by way of example only and is not intended to be limiting of the scope of the present disclosure. Also, as used in the specification including the appended claims, the singular forms “a,” “an,” and “the” include the plural, and reference to a particular numerical value includes at least that particular value, unless the context clearly dictates otherwise.

Certain terminology used in this description is for convenience only and is not limiting. The words “top”, “bottom”, “distal”, “proximal”, “inward”, “outward”, “inner”, “outer”, “above”, “below”, “axial”, “transverse”, “circumferential”, and “radial” designate directions in the drawings to which reference is made. The words “inner”, “internal”, and “interior” refer to directions towards the geometric center of the implant and/or implant adjustment tools, while the words “outer”, “external”, and “exterior” refer to direc-

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tions away from the geometric center of the implant and/or implant adjustment tools. The words, “anterior”, “posterior”, “superior”, “inferior”, “medial”, “lateral”, and related words and/or phrases are used to designate various positions and orientations in the human body to which reference is made. The term “plurality”, as used herein, means more than one. When a range of values is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. All ranges are inclusive and combinable. The terminology includes the above-listed words, derivatives thereof and words of similar import.

As used herein, the term “substantially” and derivatives thereof, and words of similar import, when used to describe a size, shape, orientation, distance, spatial relationship, or other parameter includes the stated size, shape, orientation, distance, spatial relationship, or other parameter, and can also include a range up to 10% more and up to 10% less than the stated parameter, including 5% more and 5% less, including 3% more and 3% less, including 1% more and 1% less.

An intramedullary nail is commonly secured to bone via at least one bone anchor, where each bone anchor extends directly into the surface of the bone and into a corresponding bone-anchor aperture in the intramedullary nail such that the bone anchor attaches to both the bone and the nail. However, as forces are exerted on the intramedullary nail, the attachment or bond between the bone anchor and the bone can weaken. This is especially true for patients whose bone at the bone anchor site is weakened due to osteoporosis or other bone conditions. To strengthen the attachment between the bone anchor and the bone, the bone anchor can be further secured to a bone plate that is positioned against the outer surface of the bone and secured to the bone via one or more additional bone anchors. For example, the bone plate can be positioned against the bone, and a first bone anchor can be inserted into an aperture in the plate, through the surface of the bone, and into the intramedullary nail, such that the first bone anchor attaches to the plate, the bone, and the intramedullary nail. Further, one or more other bone anchors can be inserted into the plate adjacent the first bone anchor such that the one or more other bone anchors terminate in the bone with or without passing into the intramedullary nail. The one or more other bone anchors provide additional fixation to the bone that can reduce loading on the first bone anchor.

Referring to FIG. 1, a system 10 is shown that is configured to position a bone plate 100 against a surface of a bone 70 as the bone plate 100 is fastened to the bone 70 and an intramedullary nail 60. In general, the system 10 comprises a bone plate holder 400 (e.g. a bone-plate placement tool) that supports the bone plate 100 and angulates the bone plate 100 so as to align a surface of the bone plate 100 with a surface of the bone 70. The system further comprises a bone plate adjustment tool 200 (see FIG. 8) to facilitate fixation of the bone plate 100 to the bone 70.

The system 10 can further comprise one or more of the bone plates 100, at least one bone anchor 40 such as a bone screw, an aiming assembly 50, and an intramedullary nail 60. The intramedullary nail 60 is elongate generally along a superior-inferior direction SI and is sized to be received in a medullary canal of a long bone such as a femur, tibia, or humerus. The aiming assembly 50 releasably attaches to a proximal end of the intramedullary nail 60 and can comprise an aiming guide 80 and a handle 90. The aiming assembly

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50 can facilitate alignment of the bone plate 100 such that an axis A_s of the bone plate 100 is aligned with the bone plate holder 400 and the intramedullary nail 60. For example, the axis of the bone plate 100 can be aligned with a bone-anchor opening that extends through the intramedullary nail 60.

Referring to FIGS. 2-7, the bone plate 100 includes a plate body 101 having an inner body surface 102 (e.g. a bone-facing surface) and an outer body surface 104 opposite the bone-facing surface 102 along a transverse direction T. The body-facing surface 102 can have a curved shape, contoured shape, or other shape configured to align with the surface of the bone 70. The bone plate 100 includes an outer side surface 103 that extends about an outer perimeter of the plate body 101. The outer side surface 103 can have a first transverse side 106 and a second transverse side 108 opposite from one another. The first and second transverse sides 106 and 108 can extend from the bone-facing surface 102 to the outer body surface 104. The outer side surface 103 can additionally or alternatively have a first lateral side 110 and a second lateral side 112 opposite from one another. The first and second lateral sides 110 and 112 can extend from the bone-facing surface 102 to the outer body surface 104. The first and second lateral sides 110 and 112 can extend from the first transverse side 106 to the second transverse side 108. It will be understood that embodiments of the disclosure are not limited to the specific bone plate shown in FIGS. 2-7, and that alternative bone plates are contemplated (e.g. see bone plate 300 in FIGS. 10-13).

The bone plate 100 defines a first bone-anchor aperture 116 configured to be aligned with a shaft longitudinal axis A_{S1} (see FIG. 23) when the bone plate 100 is fastened to the bone-plate placement tool 400. Thus, the longitudinal axis A_{S1} can extend through the first bone-anchor aperture 116 when the bone plate 100 is fastened to the bone plate holder 400. Further, the bone plate holder 400 is configured to align the longitudinal axis A_{S1} with the first bone-anchor aperture 116 over a full range of angles of the holder 400, without impeding with a path of a bone anchor or drill bit with the bone plate holder 400. The first bone-anchor aperture 116 can extend through the bone plate 100 from the outer body surface 104 to the bone-facing surface 102 so as to receive a bone anchor 40 (FIG. 1) to attach the bone plate 100 to the bone 70. The first bone-anchor aperture 116 can be threaded to receive a threaded head of a bone anchor. Further, the first bone-anchor aperture 116 can define variable-angle threading that permits a bone anchor to be inserted into the first bone-anchor aperture 116 at varying angles. Alternatively, the bone-anchor aperture 116 can be unthreaded.

The bone plate 100 defines at least one additional bone-anchor aperture 118, such as a plurality of additional bone-anchor apertures 118. The at least one additional bone-anchor aperture 118 is spaced from the first bone-anchor aperture 116 such that the at least one additional bone-anchor aperture 118 is offset from (i.e., not aligned with) the shaft longitudinal axis A_{S1} when the bone plate 100 is fastened to the bone-plate placement tool 20. The at least one additional bone-anchor aperture 118 extends through the bone plate 100 from the outer body surface 104 to the bone-facing surface 102. At least one of the bone-anchor apertures 118 can be threaded to receive a threaded head of a bone anchor. Further, each bone-anchor aperture 118 can define variable-angle threading that permits a bone anchor to be inserted into the bone-anchor aperture 118 at varying angles. Alternatively, each additional bone-anchor aperture 118 can be unthreaded.

The bone plate 100 defines at least one, such as a plurality of, bone-plate fasteners 114 configured to engage a fastener

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of the bone plate holder 400 so as to fasten the bone plate 100 to the holder 400. Each of the at least one bone-plate fasteners 114 can define an aperture configured to receive a corresponding fastener of the holder 400 so as to fasten the bone plate 100 to the holder 400. For example, each fastener 114 can be configured to receive a projection that extends from the bone-plate placement tool 20. It will be understood that any other suitable fasteners can be used to releasably fasten the bone plate 100 to the bone plate holder 400. For example, the fastener 114 can comprise a fixation element 350, as further described below.

The bone plate 100 further includes at least one tab member 130. The at least one tab member 130 extends from the outer side surface 103 of the bone plate 100. For example, the at least one tab 130 can extend from any one of the first and second transverse sides 106 and 108 and the first and second lateral sides 110 and 112, and/or can extend from multiple sides 106, 108, 110, and 112 of the bone plate 100. For example, FIGS. 2, 4, and 5 illustrate two tabs 130 that extend from the first lateral side 110, and a single tab 130 that extends from the second transverse side 108. Each of the at least one tabs 130 can extend in an outward direction (e.g. radial direction) from the outer side surface 103 of the bone plate 100. In an aspect, each of the at least one tabs 130 extend in a direction from the outer side surface in a direction that is offset from the shaft longitudinal axis A_{S1} when the bone plate 100 is fastened to the bone-plate placement tool 20. Each of the at least one tabs 130 can extend substantially parallel to each of the other tabs 130, can be angularly offset from each of the other tabs 130, or combinations of offset and parallel (e.g. some tabs 130 are parallel to each other, while other tabs 130 are angularly offset from other tabs 130).

Each tab 130 comprises an arm 132 and a head 134. The arm 132 extends between the plate body 101 of the bone plate 100 and the head 134. The arm 132 and the head 134 can be integrally formed as a single unitary piece. In an aspect, each tab 130 and the body 101 of the bone plate 100 can be integrally formed as a single unitary piece. Alternatively, at least one of the tabs 130 can be coupled to (e.g. attachable) to the body 101 of the bone plate 100.

Each arm 132 can extend in an outward direction from the outer side surface 103 of the plate body 101. Each arm 132 can extend substantially perpendicular to the outer side surface 103. Alternatively, each arm 132 can extend at an angle other than substantially perpendicular to the outer side surface 103. For example, with reference to FIG. 4, each arm 132 that extends from the first lateral side 110 extends in the outward direction at an angle of between 0 and 90 degrees. The arm 132a that extends from the second transverse side 108 extends in the outward direction at an angle of approximately 90 degrees. Each arm 132 can also extend in an inward or outward direction from the plate body 101. For example, with reference to FIG. 4, the arm 132a that extends from the second transverse side 108 extends at least partially in an inward direction. The position and angle of each of the tabs 130 extending from the plate body 101 can depend on, for example, the anatomy of the patient, the size and/or shape of the intramedullary nail 60, the location of corresponding apertures in the intramedullary nail 60, or other factors.

The head 134 of each tab 130 can define a tab aperture 136 that extends therethrough. With reference to FIGS. 6A and 7, the tab aperture 136a extends through the head 134 from an upper opening 138 to a lower opening 140. Each of the upper and lower openings 138 and 140 can be substantially perpendicular to a central aperture axis C_A . The central

aperture axis C_A extends through a center of the tab aperture **136a**. Alternatively, the upper and lower openings **138** and **140** can be angularly offset from a direction that is substantially perpendicular to the central aperture axis C_A . For example, the lower opening **140** can be offset from the direction substantially perpendicular to the central aperture axis C_A to align with an anatomy of the patient.

The bone-anchor aperture **136a** further includes an inner surface **142** that extends about the central aperture axis C_A . The inner surface **142** can define at least one column **144**. The at least one column **144** can extend about the central aperture axis C_A from the upper opening **138** to the lower opening **140**. Alternatively, the at least one column **144** may extend about a portion of the inner surface **142**. For example, the at least one column **144** can extend from the lower opening **140** to a location between the upper and lower openings **138** and **140**. The at least one column **144** can include threads. In an aspect, the threads of the at least one column **144** can define variable-angle threading that permits a bone anchor to be inserted into the bone-anchor aperture **136a** at varying angles. Alternatively, the at least one column **144** of the bone-anchor aperture **136a** can be unthreaded.

The inner surface **142** can further define a receiving element **146**. The receiving element **146** is configured to receive the bone plate adjustment tool **200**, as further described below. The inner surface **142** can define a first cross-sectional dimension D_1 (e.g. radial dimension extending from central aperture axis C_A), and the receiving element **146** can define a second cross-sectional dimension D_2 (e.g. radial dimension extending from central aperture axis C_A) that is greater than the first cross-sectional dimension D_1 . The receiving element **146** can comprise one or more recesses that include curved, lobed, rectangular, and/or other shapes. For example, the receiving element **146** can include a single or multi-lobe configuration, whereby each lobe of the receiving element **146** is spaced radially outward from the first cross-sectional dimension D_1 of the inner surface **142** relative to the central aperture axis C_A . The number and spacing of the lobes of the receiving element **146** can correspond to a number and spacing of a connection element **202** of the bone plate adjustment tool **200**.

The receiving element **146** can extend through the bone-anchor aperture **136a** from the upper opening **138** to the lower opening **140**. Alternatively, the receiving element **146** can extend partially through the bone-anchor aperture **136a** from the upper opening **138** to a location on the inner surface **142** between the upper and lower openings **138** and **140**. The receiving element **146** is configured to receive the bone plate adjustment tool **200** through the upper opening **138** and into the bone-anchor aperture **136a**.

The receiving element **146** can be spaced circumferentially about the inner surface **142**. For example, if the receiving element **146** includes 4 lobes, each lobe can be spaced approximately 90 degrees apart from each of the other lobes (e.g. cloverleaf configuration). Alternatively, each lobe of the receiving element **146** can be spaced at varying degrees about the inner surface, so long as the lobes correspond to the connection element **202** of the bone plate adjustment tool **200**. The receiving element **146** can also circumferentially intersect with the at least one column **144** defined by the inner surface **142**, forming peripheral areas of thread relief arranged about the inner surface **142**. For example, for a receiving element **146** including 4 lobes spaced 90 degrees apart, the at least one column **144** can be positioned within the circumferential length of the 90 degrees between each lobe. In this example, the at least one column **144** can include 4 sections spaced circumferentially

about the inner surface **142** configured to receive and engage a bone-anchor positioned within the bone-anchor aperture **136a**. If the at least one column **144** includes a threaded section, the threaded section can threadedly engage a corresponding threaded section of a bone-anchor.

Each of the bone-anchor apertures **136** defined by each tab **130** can be configured substantially similarly as each of the other bone-anchor apertures **136**. For example, the bone-anchor apertures **136** defined by the tabs **130** extending from the first lateral side **110** of the plate body **101** can have the same dimensions, threaded sections, receiving elements, or other features as the bone-anchor aperture **136a** defined by the tab **130** extending from the second transverse side **108** of the plate body **101**. Alternatively, each of the bone-anchor apertures **136** can be configured differently from one or more of the other bone-anchor apertures **136**. For example, a cross sectional dimension (e.g. diameter and/or circumference) of the bone-anchor aperture **136a** defined by the tab **130** extending from the second transverse side **108** of the plate body **101** can be greater than a cross sectional dimension of one or more of the tab apertures **136** defined by the tabs **130** extending from the first lateral side **110** of the plate body **101** and/or any of the other sides **106** and **112** of the plate body **101**. The particular configuration of the bone-anchor aperture **136** can correspond to the particular bone anchor being received within the aperture **136**.

The at least one additional bone-anchor aperture **118** and the first bone-anchor aperture **116** can be configured in a substantially similar manner as the tab apertures **136** defined by the tabs **130**. For example, at least one or more of the additional bone-anchor apertures **118** and the first bone-anchor aperture **116** can include the receiving element **146** and the at least one column **144**. Each receiving element **146** of the bone-anchor apertures **116** and **118** can be configured to correspond to the connection element **202** of the bone plate adjustment tool **200**.

Each of the tabs **130** can be configured to flex and/or bend between multiple positions. Each of the tabs **130** can transition between a pre-fixation position, in which the head **134a** is in a first position, and a fixation position, in which the head **134a** is in a second position spaced from the first position. The flexing of the tabs **130** is discussed further below with reference to FIG. **24**. The tabs **130** can comprise a grade of stainless steel or other material adapted to plastically deform. In an aspect, the tabs **130** can comprise a carburized stainless steel to avoid galvanic corrosion caused between the contact between the tab **130** and the bone-anchor **40** positioned within the bone-anchor aperture **136**. For example, the intramedullary nail **60** can comprise titanium or a titanium alloy due to its low elastic modulus that encourages bone healing. If the bone-anchor **40** is made from either a stainless steel or a titanium, there can be a risk of galvanic corrosion on one end of the anchor **40** or another, where disparate metals contact each other. In an alternative aspect, the bone-anchor **40** can comprise a carburized stainless steel to avoid galvanic corrosion. It will be appreciated that other materials can be used to form the tabs **130**, the plate body **101**, and the anchors **40**, including, for example, cobalt chrome, cobalt chrome alloys, or still other materials.

The arms **132** of each of the tabs **130** can include a substantially flat configuration. For example, each arm **132** can have a width that is greater than a thickness of the arm **132**. The flat configuration can facilitate bending, and in particular, can facilitate bending in a desired direction. For example, the arm **132** can bend in directions substantially perpendicular to the width of the arm **132**. The arm **132** can be aligned with respect to the plate body **101** of the bone

plate **100** such that the arm **132** can be bent to substantially align a contour of the bottom surface of the tab **130** with a contour of the surface of the bone **70** of the patient during a surgical procedure.

Referring to FIG. 6B, the outer body surface **104** is spaced from the inner body surface **102** a first distance L_1 along the transverse direction **T** so as to define a plate body **101** thickness. The arm **132a** defines an inner arm surface **133** configured to face the underlying bone **70**. The arm **132a** further defines an outer arm surface **135** opposite the inner arm surface **133** along the transverse direction **T**. The outer arm surface **135** is spaced from the inner arm surface **133** by a second distance L_2 in the transverse direction **T** so as to define an arm **132a** thickness. The first distance L_1 that defines the plate body **101** thickness can be greater than the second distance L_2 that defines the arm **132a** thickness.

The inner arm surface **133** is spaced from the outer arm surface **135** in an inward direction **I** that is oriented along the transverse direction **T**. In an aspect, the outer arm surface **135** is offset from the outer body surface **104** in the inward direction **I**. In an alternative or additional aspect, the outer arm surface **135** is offset with respect to the outer head surface **139** in the inward direction **I**. The inner arm surface **133** can be continuous with the inner head surface **137**. For example, the inner arm surface **133** may not be offset from the inner head surface **137** in the inward direction **I** such that the inner arm surface **133** extends adjacent to the inner head surface **137** without interruption. Alternatively, the inner arm surface **133** can be offset from the inner head surface **137** in the inward direction **I**.

Referring to FIG. 7, the arms **132a** can extend from the plate body **101** to the head **134** along a central tab axis C_T . Each arm **132a** can be configured to twist about the central axis C_T so as to move the head **134** from the pre-fixation position to the fixation position. Each arm **132** can include a first taper portion **129** and a second taper portion **131**. The first taper section **129** is located adjacent to the body **101**. The outer arm surface **135** within the first taper section **129** can taper in an outward direction. The outward direction being opposite the inward direction **I**. The inner arm surface **133** within the first taper section **129** can taper in the inward direction **I**. For example, the thickness of the arm **132a** within the first taper section **129** can be greater than a thickness of the arm **132** along a length of the arm between the first taper section **129** and the second taper section **131**.

The second taper section **131** is located adjacent to the head **134a**. The outer arm surface **135** within the second taper section **131** can taper in an outward direction. The inner arm surface **133** within the second taper section **131** can taper in the inward direction **I**. For example, the thickness of the arm **132a** within the second taper section **131** can be greater than a thickness of the arm **132a** along a length of the arm between the first taper section **129** and the second taper section **131**.

Alternatively, or additionally, a width of the arm **132a** within the first taper section **129** and the second taper section **131** can increase along its length from the arm **132a** toward the body **101** and along its length from the arm **132a** toward the head **134a**, respectively. The width of the arm **132a** defining a dimension that is substantially perpendicular to the inward direction **I** and the length of the arm **132a** between the first and second taper sections **129** and **131**. The width of the arm **132a** within the first taper section **129** can be greater than a width of the arm **132a** along its length between the first and second taper sections **129** and **131**, and a width of the arm **132a** within the second taper section **131**

can be greater than a width of the arm **132a** along its length between the first and second taper sections **129** and **131**.

The head **134a** defines an inner head surface **137** configured to face the underlying bone, and an outer head surface **139** opposite the inner head surface **137**. The outer head surface **139** is spaced from the inner head surface **137** by a third distance L_3 so as to define a head **134a** thickness. In an aspect, the third distance L_3 defining the head **134a** thickness is substantially equal to the first distance L_1 defining the plate body **101** thickness. In another aspect, the third distance L_3 defining the head **134a** thickness is substantially equal to the second distance L_2 that defines the arm **132a** thickness. In an aspect, the thicknesses of the arm **132a**, the head **134a**, and the plate body **101** can be different.

Referring to FIGS. 8 and 9, the bone plate adjustment tool **200** includes the connection element **202**, a shaft **204**, and a handle **206**. The connection element **202** is positioned at the distal end **201** of the adjustment tool **200**. The shaft **204** extends between the connection element **202** and the handle **206**. In an aspect, the connection element **202** extends distally from a distal most end of the shaft **204**. The handle **206** extends from the shaft **204** toward a proximal end **203** of the adjustment tool **200**. The handle **206**, the shaft **204**, and the connection element **202** can be rigidly connected together such that movement and/or rotation of the handle **206** causes movement and/or rotation of the connection element **202**. The bone plate adjustment tool **200** is configured to engage and transition the tabs **130** from the pre-fixation position to the fixation position.

The connection element **202** can comprise one or more protrusions **210** that include curved, lobed, rectangular, and/or other shapes. For example, the connection element **202** can include a single or multi-lobe configuration, whereby each lobe (e.g. protrusion **210**) of the connection element **202** protrudes in a radially outward direction. Each lobe of a multi-lobe configuration can be spaced circumferentially about a center of the connection element **202**. The number and spacing of the protrusions **210** of the connection element **202** can correspond to a number and spacing of recesses of the receiving element **146** defined by the bone plate **100**.

The connection element **202** is configured to engage the receiving element **146** to at least temporarily connect the adjustment tool **200** to the bone plate **100**. The connection between the adjustment tool **200** and the bone plate **100** can enable the adjustment tool to move and/or manipulate the bone plate **100**. For example, the adjustment tool **200** can bend and/or flex the tabs **130**. Additionally, the configurations of the receiving element **146** and the connection element **202** can reduce the risk of damage to the inner surface **142** of the bone-anchor aperture **136**. For example, if the inner surface **142** includes the at least one column **144** having a threaded region, the configurations of the receiving element **146** and the connection element **202** can reduce the risk of thread damage.

It will be appreciated that more than one bone plate adjustment tools **200** can connect with the bone plate **100**. For example, a bone plate **100** having more than one receiving element **146** that have different configurations (e.g. size, shape, etc.), multiple bone plate adjustment tools **200** can be used. In another example, a first bone plate adjustment tool **200** can engage a first bone-anchor aperture **136a**, and a second bone plate adjustment tool **200** can engage a second bone-anchor aperture **136a**. The first bone plate adjustment tool **200** can be used to, for example, transition the tab **130** while the second bone plate adjustment tool **200** provides, for example, a counter torque to maintain the

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position of the bone plate **100**. Each bone plate adjustment tool **200** can comprise a connection element **202** having a configuration (e.g. size, shape, etc.) that corresponds to a configuration of the respective receiving element **146**.

FIGS. **10-13** illustrate an alternate aspect of a bone plate **300**, according to aspects of this disclosure. Portions of the alternate aspect of the bone plate **300** disclosed in FIGS. **10-13** are similar to aspects of the bone plate **100** described above in FIGS. **2-7** and those portions function similarly to those described above. The bone plate **300** includes a body **301** having a bone-facing surface **302** and an outer surface **304** opposite the bone-facing surface **302**. The bone plate **300** includes an outer side surface **303** that extends about an outer perimeter of the body **301**.

The bone plate **300** defines a first bone-anchor aperture **316** that extends through the bone plate **300** from the outer surface **304** to the bone-facing surface **302** so as to receive the bone anchor **40** to attach the bone plate **300** to the bone **70**. The bone plate **300** defines at least one additional bone-anchor aperture **318**, such as a plurality of additional bone-anchor apertures **318**. The at least one additional bone-anchor aperture **318** is spaced from the first bone-anchor aperture **316**. The bone plate **300** further includes at least one tab member **330**.

The bone plate **300** further includes the fixation element **350**. The fixation element **350** extends at least partially through the outer surface **304** and includes a recessed portion **352**, a first aperture **354**, and a second aperture **356** (e.g. a plate fixation element). The fixation element **350** is configured to receive a bone plate holder **400** within to connect the bone plate **300** to the bone plate holder **400**. The fixation element **350** can include fewer or more apertures **354** and **356** to align with corresponding elements of the bone plate holder **400**.

The recessed portion **352** extends into the body **301** from the outer surface **304** to an inner recessed surface **358**. The first and second apertures **354** and **356** both extend from the inner recessed surface **358** into the body **301** toward the bone-facing surface **302**. In an aspect, one or both of the first and second apertures **354** and **356** extend through the body **301** to the bone-facing surface **302**. The first aperture **354** can be threaded to receive a corresponding threaded portion of a plate coupling element **432** of the bone plate holder **400**. The threads defined by the first aperture **354** can extend along an entire length of the first aperture **354**. Alternatively, the threads can extend along a partial length of the first aperture **354**. For example, the threads can extend from the inner recessed surface **358** toward the bone-facing surface **302**.

The second aperture **356** is spaced apart from the first aperture **354** within the recessed portion **352**. The second aperture **356** can be unthreaded and configured to receive a corresponding portion (e.g. a holder fixation element **422**) of the bone plate holder **400**. The second aperture **356** can have a cylindrical shape, a conical shape, or other shape that corresponds to the holder fixation element **422** of the bone plate holder **400**.

In an alternative aspect, one or both of the apertures **354** and **356** can be positioned within the recessed portion **352**. For example, one or both of the apertures **354** and **356** can extend into the body **301** from the outer surface **304** toward the bone-facing surface **302**.

The recessed portion **352** includes an inner edge **353** extending about a perimeter of the recessed portion **352**. The inner edge **353** has a first end **355** spaced from a second end **357** in a plate lateral direction **L**. The second aperture **356** is positioned between the first aperture **354** and the first end

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355 in the plate lateral direction **L**. The first aperture **354** is positioned between the second end **357** and the second aperture **356** in the lateral direction **L**. The inner edge **353** tapers in width, as measured in a direction substantially perpendicular to the plate lateral direction **L**, from a maximum width W_1 of the inner edge **353** located toward the second end **357** to a minimum width W_2 located toward the first end **355**. A length between the maximum width W_1 and the minimum width W_2 along the inner edge **353** can be substantially linear.

It will be appreciated that the fixation element **350** can include additional elements, for example, additional apertures, retentions elements, recesses/protrusions, or other elements. The fixation element **350** can also include other fixation configurations to connect to the bone plate holder **400**. For example, the fixation element **350** can include a snap-fit, friction-fit, or other type of fixation configuration.

FIGS. **14-20** illustrate the bone plate holder **400**, according to aspects of this disclosure. The bone plate holder **400** includes a body **402** (e.g. holder body), a shaft **404**, and a grip member **406**. The grip member **406** is configured to be gripped by a surgeon performing a surgical procedure to align the bone plate holder **400** with the bone plate **100**, and to control the position of the bone plate **100** while the bone plate **100** is being fastened to the intramedullary nail **60**. The grip member **406** can connect to a handle connection portion **403** of the body **402**. The body **402** and the shaft **404** are configured to control connecting and disconnecting the bone plate holder **400** with the bone plate **100**.

With reference to FIGS. **15** and **16**, the body **402** has an inner surface **410** that extends through the body **402** from a proximal surface **411** at a proximal end **412** of the body **402** to a distal surface **413** at a distal end **414** of the body **402**. The proximal end **412** is spaced from the distal end **414** in a proximal direction **P**. The inner surface **410** defines a shaft channel **416** that extends along a central channel axis C_s through the body **402** from the proximal end **412** to the distal end **414**. The inner surface **410** further defines a shaft retention portion **418**. The shaft retention portion **418** is located toward the proximal end **412** of the body **402**. The shaft retention portion **418** can include a locking feature **420** to at least temporarily secure the shaft **404** within the shaft channel **416**. In an aspect, the locking feature **420** can include a threaded portion configured to engage with a corresponding body coupling element **434** of the shaft **404**.

The body **402** further includes the holder fixation element **422** (e.g. a projection) configured to interface with the second aperture **356** of the bone plate **300**. The holder fixation element **422** can extend distally from the distal surface **413**. The holder fixation element **422** can have a substantially cylindrical shape, conical shape, or other shape. The shape and/or configuration of the holder fixation element **422** can correspond to a shape and/or configuration of the second aperture **356** of the bone plate **300**. The interface between the holder fixation element **422** and the second aperture **356** facilitates alignment of the bone plate holder **400** with the bone plate **300**. It will be appreciated that the body **402** can include multiple holder fixation elements **422**, and the bone plate **300** can include multiple second apertures **356** that correspond to the multiple holder fixation elements **422**.

Referring to FIG. **17**, the distal surface **413** at the distal end **414** of the body **402** includes an outer edge **415** extending about a perimeter of the distal end **414**. The outer edge **415** has a first end **417** spaced from a second end **419** in a lateral holder direction **L'**. The fixation element **422** is positioned between a distal opening **421** of the shaft channel

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416 defined by the distal surface 413 and the first end 417 in the lateral holder direction L'. The distal opening 421 is positioned between the second end 419 and the fixation element 422 in the lateral holder direction L'. The outer edge 415 tapers in width, as measured in a direction substantially perpendicular to the lateral holder direction L', from a maximum width W'_1 of the outer edge 415 located toward the second end 419 to a minimum width W'_2 located toward the first end 417. A length between the maximum width W'_1 and the minimum width W'_2 along the outer edge 415 can be substantially linear.

A configuration of the distal end 414 of the body 402 can be configured to correspond to the recessed portion 352 of the fixation element 350 of the bone plate 300 so as to align the bone plate holder 400 with the bone plate 300 during connection of the holder 400 with the plate 300. For example, the distal end 414 of the body 402 can be received within the recessed portion 352 such that the plate lateral direction L aligns with the lateral holder direction L' to at least partially rotationally lock the holder 400 with the plate 300 while the shaft 404 of the holder 400 is being secured to the fixation element 350 of the plate 300. The maximum width W'_2 and the minimum width W'_1 of the outer edge 415 of the body 402 can be slightly less than the maximum width W_2 and the minimum width W_1 of the recessed portion 352, respectively, such that an outer surface 405 of the body 402 contacts an inner surface 351 of the recessed portion 352. The distal end 414 of the body 402 can be inserted into the recessed portion 352 of the plate 300 such that the distal surface 413 is substantially flush against the inner recessed surface 358.

With reference to FIG. 18, the shaft 404 includes a control element 430 (e.g. a shaft grip member), the plate coupling element 432, and the body coupling element 434. The control element 430 is located at a proximal end 436 of the shaft 404. The plate coupling element 432 is located at a distal end 438 of the shaft 404. The body coupling element 434 is located between the plate coupling element 432 and the control element 430 along the shaft 404. The shaft 404 is configured to be inserted into the shaft channel 416 of the body 402 through the proximal end 412. The control element 430 can be positioned externally to the shaft channel 416, such that the control element 430 extends out of the proximal surface 411 of the body 402 in the proximal direction P and is accessible by a surgeon during a procedure. A distal surface of the control element 430 can abut against the proximal surface 411 of the body 402 when the shaft 404 is positioned within the shaft channel 416.

The plate coupling element 432 can extend through an opening defined by the distal end 414 of the body 402, such that the plate coupling element 432 extends distally out from the distal surface 413 of the body 402.

The body coupling element 434 is configured to engage with the locking feature 420 of the body 402. In an aspect, the body coupling element 434 can include a threaded portion configured to engage with a corresponding threaded portion of the locking feature 420. For example, during insertion of the shaft 404 into the shaft channel 416 of the body 402, the body coupling element 434 can engage the locking feature 420 until the body coupling element 434 is positioned distal to the locking feature 420. The position of the body coupling element 434 distal to the locking feature 420 can provide at least a temporary lock between the shaft 404 and the body 402, such that the shaft 404 is at least partially restricted from exiting the shaft channel 416 through the proximal end 412 of the body 402. For example, when the locking feature 420 and the body coupling element

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434 are engaged and as the shaft 404 is rotated in a first direction of rotation (e.g. a clockwise direction), the body coupling element 434 moves distal to the locking feature 420 within the shaft retention portion 418. When the body coupling element 434 is in the position distal of the locking feature 420 of the body 402, the shaft 404 is then rotatable in the first direction of rotation without translating relative to the body 402.

The plate coupling element 432 is configured to connect with the first aperture 354 of the fixation element 350 of the bone plate 300. The plate coupling element 432 can include a threaded portion 440 that engages a corresponding threaded section of the first aperture 354. The shaft 404 can be rotated by rotating the control element 430 relative to the body 402 to threadedly connect the holder 400 to the bone plate 300. When the plate coupling element 432 is connected to the first aperture 354, and the holder fixation element 422 is connected to the second aperture 356, the bone plate holder 400 is rigidly received within the plate 300 such that linear and rotational movement between the holder 400 and the plate 300 is substantially prevented. The rigid connection between the holder 400 and the plate 300 allows the surgeon to control the movement and placement of the bone plate 300 relative to the bone 70 of the patient during a surgical procedure.

After the plate 300 is secured to the bone 70, the bone plate holder 400 can be removed from the bone plate 300. The shaft 404 can be disconnected from the first aperture 354 by rotating the control element 430 in a direction opposite to the direction of rotation for securing the plate coupling element 432 to the first aperture 354. After the plate coupling element 432 is disconnected, the holder 400 can be removed by retracting the distal end 414 of the body 402 from the recessed portion 352 and by retracting the holder fixation element 422 from the second aperture 356.

The control element 430 can include an engagement element 442 that defines a hexagonal or any alternatively shaped structure that can be engaged by a screw driving instrument, a protrusion element engageable by a wrench, or a cross hole configured to receive a cylindrical bar to rotate the control element 430 the bone plate holder 400 is being connected and disconnected to the plate 300.

With reference to FIGS. 21-26, an example embodiment of a method of implanting an intramedullary nail 60 will be described. As shown in FIG. 21, the handle 90 is coupled to the intramedullary nail 60, and the intramedullary nail 60 is driven into the medullary canal of the bone 70 by the handle 90. In FIG. 22, the aiming guide 80 of the aiming assembly 50 is fastened to the handle 90. The bone plate 300 is fastened to the bone plate holder 400 such that the bone plate 300 is configured to angulate with the plate holder 400 to position the bone plate 300 on the bone 70. Although reference is made below with respect to bone plate 300, it will be appreciated that the description also applies to bone plate 100 and/or other bone plate configurations.

The patient can be oriented so that anatomic lateral shows condylar overlap (e.g. perfect anatomic lateral), and the nail 60 can be rotated to a desired position (e.g. obtain perfect circles). The nail 60 can be locked to the bone 70 (e.g. distal fragment) with a bone screw. In an aspect, the initial screw to lock the nail 60 to the bone 70 is a medial oblique screw. The screw substantially prevents motion of the nail 60 relative to the bone 70.

With reference to FIG. 23, an incision can be made and the bone plate 300 is introduced to the bone 70. The incision can include, for example, an 8 cm incision to connect first and second lateral guide sleeves 92 and 94. The first guide

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sleeve **92** can extend along the shaft longitudinal axis A_{S1} , and the second guide sleeve **94** can extend along a second longitudinal axis A_{S2} . The first guide sleeve **92** aligns with the first bone-anchor aperture **316** and a first aperture in the intramedullary nail **60**, and the second guide sleeve **94** aligns with one of the additional bone-anchor apertures **318** and a second aperture in the nail **60**. A first drill bit (not shown) can be inserted into one of the first and second guide sleeves **92** and **94** to drill a first hole in the bone **70**. A second drill bit (not shown) can be inserted into the other of the first and second guide sleeves **92** and **94** to drill a second hole in the bone **70**. One of the first and second drill bits is removed from the bone, and a first bone-anchor screw (e.g. 5.0 VA locking screw) is inserted into the hole. The first bone-anchor can be positioned such that the bone-anchor is approximately 1 cm proud. After the first bone-anchor has been inserted, a second bone-anchor screw (e.g. 5.0 VA locking screw) is inserted into the other hole formed. The second bone-anchor can be positioned such that the bone-anchor is approximately 1 cm proud.

After the first and second bone-anchors have been inserted, a lateral oblique bone-anchor screw can be inserted. After the first and second bone-anchors and the lateral oblique screw have been inserted, a lateral pressure can be applied to the plate **300** so the center and anterior portions of the plate **300** fit to the bone **70**. This can push one or more of the tabs **330** off the bone **70**. After the plate **300** is fit to the bone **70**, the first and second bone-anchors are tightened. The bone plate holder **400** can be removed from the plate **300** by rotating the control element **430** to release the plate coupling element **432** of the shaft holder **400** from the first aperture **354** of the bone plate **300**. The anterior locking screws can be inserted through the additional bone-anchor apertures **318**. In an aspect, holes can be pre-drilled in the bone prior to inserting the anterior locking screws.

With reference to FIG. **24**, the bone plate adjustment tool **200** can manipulate each of the at least one tab members **330** so that the posterior portions of the plate **300** fit to the bone **70**. For example, the connection element **202** of the adjustment tool **200** is inserted into the respective aperture **336** of the tab **330**, and a pressure (e.g. a deflection force) can be applied to the respective head **334** of the tab **330** to cause the arm **332** to deflect the head **334** from the pre-fixation position to the fixation position. For example, the arm **332** can deflect with respect to the plate body **301** in response to a force so as to move the head **334** between the pre-fixation position and the fixation position. The force applied to the arm **332** to transition the head between the pre-fixation position and the fixation position can be insufficient to cause the plate body **101** to deflect. After the head **334** is transitioned to the fixation position, the inner head surface **137** abuts the underlying bone. In an aspect, the arm **332** is deformable such that the fixation position of the head **334** is offset in the inward direction **I** with respect to the pre-fixation position of the head **334**.

During transition, the trajectory of the screw through the respective aperture **336** should be considered to ensure the screw is securely located within the bone **70**. After the tabs **330** have been transitioned to their respective fixation position, posterior bone-anchor screws can be inserted through each aperture **336** and into the bone **70**. After each of the bone-anchors has been inserted, a final torque can be applied to the bone-anchors.

Although the disclosure has been described in detail, it should be understood that various changes, substitutions, and alterations can be made herein without departing from the spirit and scope of the invention as defined by the

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appended claims. Additionally, any of the embodiments disclosed herein can incorporate features disclosed with respect to any of the other embodiments disclosed herein. Moreover, the scope of the present disclosure is not intended to be limited to the particular embodiments described in the specification. As one of ordinary skill in the art will readily appreciate from that processes, machines, manufacture, composition of matter, means, methods, or steps, presently existing or later to be developed that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present disclosure.

What is claimed:

1. A system comprising:

a bone plate including:

a plate body that defines a bone-facing body surface configured to face a bone and an outer body surface opposite the bone-facing body surface, wherein the bone plate defines a plurality of threaded bone anchor apertures that extend through the plate body from the outer body surface to the bone-facing body surface and are each configured to receive a respective bone anchor, the plate body is devoid of flexible arms, and the plate body defines an outermost plate body perimeter that extends about the plate body at a location between the bone-facing body surface and the outer body surface; first and second tabs each including a respective flexible arm that extends from the plate body along a respective length direction to a respective head that is supported by the respective flexible arm, and a respective tab aperture that extends through the respective head and is configured to receive a respective bone anchor, wherein each of the first and second tabs terminates at the respective head,

wherein each of the flexible arms defines a respective inner bone-facing surface and a respective outer surface opposite the respective inner bone-facing surface along a thickness that extends along a thickness direction,

wherein each of the flexible arms defines respective opposed sides that are opposite each other along a width that extends along a width direction, and the width is greater than the thickness along respective entireties of the flexible arms, and the sides are all straight and linear,

wherein the opposed sides of the flexible arm of the first tab are spaced from each other along the outermost body perimeter along the width direction, and the opposed sides of the flexible arm of the second tab are spaced from each other along the outermost body perimeter along the width direction,

wherein both of the respective opposed sides of the first flexible arm extend directionally away from both of the respective opposed sides of the second flexible arm from the plate body to the respective head; and an intramedullary nail configured to be inserted into medullary canal of the bone, wherein the intramedullary nail is configured to receive the bone anchors so as to fix the intramedullary nail to the plate body.

2. The system of claim 1, wherein each of the threaded bone anchor apertures defines respective pairs with each other of the threaded bone anchor apertures, and for each respective pair of threaded bone anchor apertures, a straight line that extends through respective centers of the bone anchor apertures of the pair divides the plate body into a first plate body side and a second plate body side, and the tabs

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extend from only one of the first and second plate body sides, such that no tabs extend out from the other of the first and second plate body sides.

3. The system of claim 1, wherein:

the plate body defines a first direction, a second direction 5 perpendicular to the first direction, and a third direction opposite the second direction and perpendicular to the first direction,

the bone plate defines a bone plate aperture,

wherein the bone plate aperture is disposed between first 10 and second ones of the threaded bone anchor apertures with respect to the first direction, such that a straight line that passes through the bone plate aperture and is oriented along the first direction also passes through the 15 first and second ones of the threaded bone anchor apertures,

wherein the plurality of threaded bone anchor apertures further includes third and fourth ones of the plurality of threaded bone anchor apertures that are offset with 20 respect to the bone plate aperture in the second direction, such that a second straight line that is oriented along the first direction and extends through the third one of the plurality of fixation holes also passes through the fourth one of the plurality of fixation holes; and 25 each of the respective heads is offset from the plate body in the third direction.

4. The system of claim 1, wherein the threaded bone anchor apertures comprise variable angle threading.

5. The system of claim 1, wherein the bone plate further 30 comprises a bone plate aperture that extends through the plate body that is configured to receive a corresponding fastener of a bone plate holder.

6. The system of claim 5, wherein the bone plate aperture 35 is threaded.

7. The system of claim 6, further comprising a second bone plate aperture that is configured to receive a respective feature of the bone plate holder.

8. The system of claim 7, wherein the second bone plate 40 aperture is unthreaded.

9. The system of claim 1, further comprising the bone plate holder.

10. The system of claim 1, wherein the plate body has a plate body thickness that extends from the bone-facing body surface to the outer body surface, and an entirety of the plate 45 body thickness is greater than the thickness of each of the flexible arms.

11. A system comprising:

a bone plate including:

a plate body that defines a bone-facing body surface 50 configured to face a bone and an outer body surface opposite the bone-facing body surface, wherein the bone plate defines a plurality of threaded bone anchor apertures that extend through the plate body from the outer body surface to the bone-facing body surface and are each configured to receive a respective bone anchor, and the plate body is devoid of flexible arms; and

first and second tabs each including a respective flexible arm that extends from the plate body, a respective head supported by the respective flexible arm, and a respective tab aperture that extends through the 60 respective head and is configured to receive a respective bone anchor,

wherein each of the flexible arms defines a respective 65 inner bone-facing surface and a respective outer surface opposite the respective inner bone-facing

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surface along a respective thickness that extends along a thickness direction, and

wherein each of the flexible arms defines respective opposed sides that are opposite each other along a width that extends along a width direction, and the width is greater than the thickness along respective entireties of the flexible arms, and

wherein both of the respective opposed sides of each of the first and second flexible arms extend directionally away, along respective entireties of their lengths from the plate body to the respective head, from the other of the first and second flexible arms; and

an intramedullary nail configured to be inserted into medullary canal of the bone, wherein the intramedullary nail is configured to receive the bone anchors so as to fix the intramedullary nail to the plate body.

12. The system of claim 11, wherein each of the threaded bone anchor apertures defines respective pairs with each other of the threaded bone anchor apertures, and for each respective pair of threaded bone anchor apertures, a straight line that extends through respective centers of the bone anchor apertures of the pair divides the plate body into a first plate body side and a second plate body side, and the tabs extend from only one of the first and second plate body sides, such that no tabs extend out from the other of the first and second plate body sides.

13. The system of claim 11, wherein:

the plate body defines a first direction, a second direction perpendicular to the first direction, and a third direction opposite the second direction and perpendicular to the first direction,

the bone plate defines a bone plate aperture,

wherein the bone plate aperture is disposed between first and second ones of the threaded bone anchor apertures with respect to the first direction, such that a straight line that passes through the bone plate aperture and is oriented along the first direction also passes through the first and second ones of the threaded bone anchor apertures,

wherein the plurality of threaded bone anchor apertures further includes third and fourth ones of the plurality of threaded bone anchor apertures that are offset with respect to the bone plate aperture in the second direction, such that a second line that is oriented along the first direction and extends through the third one of the plurality of fixation holes also passes through the fourth one of the plurality of fixation holes; and

each of the respective heads is offset from the plate body in the third direction.

14. The system of claim 11, wherein the bone plate further comprises a bone plate aperture that extends through the plate body that is configured to receive a corresponding fastener of a bone plate holder.

15. The system of claim 14, wherein the bone plate aperture is threaded.

16. The system of claim 15, further comprising a second bone plate aperture that is configured to receive a respective feature of the bone plate holder.

17. The system of claim 16, wherein the second bone plate aperture is unthreaded.

18. The system of claim 11, further comprising the bone plate holder.

19. The system of claim 11, wherein the opposed sides of the first and second flexible arms extend straight and linearly from the plate body to the respective heads.

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20. The system of claim 11, wherein the plate body defines an outermost plate body perimeter that extends about the plate body at a location between the bone-facing body surface and the outer body surface, and the width direction extends along the outermost plate body perimeter at an interface between the outermost plate body perimeter and the arms. 5

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